

## PHYSICAL SECURITY : CONCISE NOTES

### Chapter – 1 : Concepts in Risk Management

#### 1. Enterprise Security Risk Management (ESRM)

- 1.1. **Strategic approach** to Security management, that ties an organizations security practice to its overall strategy using globally accepted and established risk management principles.
- 1.2. The objective of ESRM is to **identify, evaluate, and mitigate** the impact of security risks to the organization with priority given to protective activities that help enable the organization to advance its overall mission.
- 1.3. ESRM is a **management** process **NOT** a program.

#### 2. How is ESRM Different.

| <b>Traditional Risk Management</b>                                                                                                                              | <b>ESRM</b>                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Built around a <b>security program</b> with numerous sub-programs                                                                                               | ESRM <b>addresses all security risks</b> to an organisations assets and developing specific mitigation steps.                                                                  |
| Addresses a <b>specific threat</b> or issue by a program.                                                                                                       | ESRM does not attempt to <b>assign a risk</b> to a specific program.                                                                                                           |
| In the traditional security management process, the <b>security professional does what is requested</b> or directed and implements security processes/programs. | In ESRM, <b>asset owners own decisions</b> for the risk to the assets they manage. Those decisions are made with the input and <b>guidance from the security professional.</b> |

#### 3. Benefits of ESRM. ESRM provides many benefits to the organization.

- 3.1. Enabling critical decisions around security risks at an enterprise level. **supporting** the **organization's mission and objectives.**
- 3.2. Allowing asset owners and other stakeholders to develop a greater. consistent **understanding** of the security function's role.
- 3.3. Providing a **mechanism** to elevate identified security risks to top management.
- 3.4. Better **alignment** of security resources and organizational strategy to effectively manage prioritized risk.
- 3.5. Improving **effectiveness** and efficiency of the security program.
- 3.6. Early **identification** and proactive **monitoring** of threats and vulnerabilities.
- 3.7. More effective risk **prioritization** and mitigation based on partnership between security and asset owners.
- 3.8. Improving **engagement** with stakeholders with a vested interest in the security of the organization.
- 3.9. Better support for the organization's legal. regulatory. contractual. and internal audit responsibilities.
- 3.10. Integrating security into the culture of the organization.

#### 4. Components of ESRM. ESRM has three primary components:

- 4.1. The **context of ESRM**, which includes organizational aspects that security professionals must understand to successfully adopt ESRM;

- 4.2. The **ESRM cycle**, which is ESRM's actual process of security risk management that emphasizes the importance of understanding assets; and
- 4.3. The **foundation of ESRM**, which includes organizational concepts that support the ESRM approach and maximize its impact.

## **5. The Context of ESRM.**

5.1. When implementing ESRM, security professionals should have a understanding of the organization's:

- 5.1.1. **Mission** and vision
- 5.1.2. Core **values**
- 5.1.3. Operating **environment**
- 5.1.4. **Stakeholders**.

### **5.2. Mission & Vision.**

5.2.1. Understanding of the organization's mission and vision is critical to effectively identify risks that could impede the organization's ability to achieve its goals and objectives.

5.2.2. Security professionals need to consider the following aspects such as.

- Operating **structure**
- Key staff and **leadership**
- **Products** and services
- **Regulations** and legal requirements
- **Long-term** strategic goals and objectives

### **5.3. Core Values.**

5.3.1. Organizational **core values** can indicate how well ESRM and its emphasis on partnership, collaboration and **transparency** might be embraced and supported by the corporate culture.

5.3.2. Core values may include things like **environmental** stewardship, the community, employee safety and security, product quality, brand and image protection and new product development.

5.3.3. Linking ESRM to the organization's core values will better ensure alignment with the priorities of top management.

5.3.4. An organization's core values are an integral component and driver of corporate culture.

### **5.4. Operating Environment.**

5.4.1. To assess risk, it is essential that security professionals understand the environment in which the organization operates.

5.4.2. The operating environment consists of three main categories:

- **Physical**: Includes factors such as the organization's types and locations of facilities or campuses.
- **Non physical**: Includes factors such as the **geo-political landscape**: culture: industry pressures: legal. regulatory and compliance requirements: intensity of competition: organizational growth mode: speed of decision making: and willingness to adopt technology.
- **Logical**: Logical factors include the organization's various information types. Also included are **digital assets and the network or digital space** that connects them to each other and to stakeholders.

## 5.5. Stakeholders.

5.5.1. Any individual who **interfaces** with the organization may be considered a stakeholder.

5.5.2. Security professionals should know stakeholders of the organization and understand what is important to those stakeholders.

5.5.3. Examples of stakeholders include:

- Persons responsible for defining the organization's strategy
- Organizational owners of tangible and intangible assets
- Persons who are working for or on behalf of the organization
- Persons who contribute knowledge or support to the organization: el Clients/constituents of the organization:
- Surrounding communities impacted by the organization.

5.5.4. Understanding stakeholders does not necessarily mean harmonizing their interests, but rather understanding their needs and their risk insights to better execute the ESRM cycle.

## 6. ESRM Cycle.

6.1. The ESRM Cycle is the part of the ESRM approach that describes how security risks are to be mitigated.

6.2. Its most defining characteristic is its emphasis on understanding organizational assets and involving asset owners in the risk management process.



6.3. The **ESRM Cycle** includes four processes:

- Identify and prioritize **assets**
- Identify and prioritize **risks**
- **Mitigate** prioritized risks

- **Continuous** improvement.

#### 6.4. **Step – I : Identify & Prioritize Assets.**

6.4.1. **Asset prioritization** is based on each asset's criticality to the organization's mission and overall strategy. Based on inputs from asset owners and other stakeholders.

6.4.2. The resulting prioritized lists of assets is the basis for the remainder of the ESRM Cycle.

6.4.3. Key roles to understand throughout the ESRM Cycle include:

- **Asset owner**

- Person most directly responsible for successful operation of the asset.
- **ESRM assigns responsibility for the risk to an asset to the asset owner.**
- The asset owner is the risk owner.

- **Top management.** Highest level of executive leadership in an organization.

- **Security professional.** Acts as a security risk subject matter expert and a trusted advisor to asset owners, top management, and other stakeholders.

#### 6.5. **Step – II : Identify & Prioritize Risks.**

6.5.1. Risk prioritization is based on each risk's potential to undermine the organization's ability to execute its mission and overall strategy.

6.5.2. Risk prioritization is based in part on guidance from top management and input from other stakeholders.

6.5.3. 'The resulting prioritized lists of risks & prioritized list of assets, form a set of strategic priorities throughout the entire ESRM cycle and help guide asset owner decisions.

#### 6.6. **Step – III : Mitigate Prioritized Risks.**

6.6.1. Risks are mitigated in order of priority, using security controls recommended by the security professional and approved by the asset owner.

6.6.2. Asset owners make the decisions with guidance from the security professional.

6.6.3. **The plan is ultimately executed by the security professional.**

#### 6.7. **Step – IV : Continuous Improvement.**

6.7.1. Spans the entire ESRM Cycle.

6.7.2. Continuous improvement of the security program occurs because of ESRM outcomes such as improved communication, greater visibility into security risks, and effective risk prioritization and mitigation.

### 7. **Foundation of ESRM.**

7.1. To ensure sustained longevity and success, there are four other critical concepts to incorporate. Called the Foundation of ESRM, its four components are:

7.1.1. **Holistic Risk Management**

7.1.2. **Partnership** with Stakeholders

7.1.3. **Transparency**

7.1.4. **Governance.**

#### 7.2. **Holistic Risk Management.**

7.2.1. The term "holistic" refers to an inclusive, all-encompassing, or complete approach.

7.2.2. The concept of holistic risk management has two relevant meanings from the perspective of ESRM.

- Participation of all stakeholders.
- Account for all types of security risk without qualifiers or silos such as physical, cyber, information, or personal security.

**7.3. Partner with Stakeholders.** ESRM encourages security professionals to identify, engage, and align with stakeholders in the security program and the organization.

**7.4. Transparency.**

7.4.1. The term transparency refers to a clear, open, and honest approach.

7.4.2. Two types of transparency are relevant from the perspective of ESRM.

- **Risk Transparency:** This suggests that security professionals should represent risks based on their understanding and expertise in a clear, open, and honest way. Security professionals are encouraged to use objective statements, quantitative analysis, and quantifiable metrics as much as possible.

- **Process Transparency:** This suggests that security professionals should ensure that asset owners and stakeholders understand the organization's security risk management process and each step of that process as security professionals guide them through it.

**7.5. Governance.**

7.5.1. Governance refers to the rules and processes by which a function or organization is governed.

7.5.2. Governance also ensures that efforts across the organization ultimately satisfy the needs of the organization.

7.5.3. Two types of governance are relevant from the perspective of ESRM.

- **Organizational Governance:** This is the system by which an organization is directed and controlled.

- **ESRM Governance:** This is the process of setting enterprise security risk policy and direction, allocating resources, and ensuring compliance, ESRM governance is a subset of corporate governance and is modeled after organizational governance.

**ESRM governance is carried out by the organization's security governance body**  
ESRM Governance body members should mostly be **security risk stakeholders, not security professionals.**

## Chapter - 2 : Functions of Physical Security

### 1. Definition & Purpose of Physical Security.

1.1. Physical Security. That part of security concerned with physical measures designed to **safeguard people**; to prevent unauthorized access to equipment, facilities, **material**, and **documents**; and to safeguard them against a security incident.

1.2. Security: The condition of being protected against hazards, threats, risks, tampering, or loss.

1.3. Physical security measures must be considered in the context of their ability to **deter an adversary**, **detect an attack**, **delay an attack**, and **deny** an adversary access to the target.

1.4. The concept of the **four Ds** should form the basis for all physical security projects.

### 2. Functions of Physical Security.

2.1. **Access control** relates to any technique used to limit or otherwise manage access to an area, facility, compound, system, person, or other asset.



2.2. **Deterrence** refers to barriers, technology, and procedures taken to convince a potential attacker that they will not be successful, and therefore influence them to avoid the site. **Of all the security measures this one is the most difficult to measure.**

2.3. **Detection** is the capacity of the physical protection system to sense that a potential intruder is present.

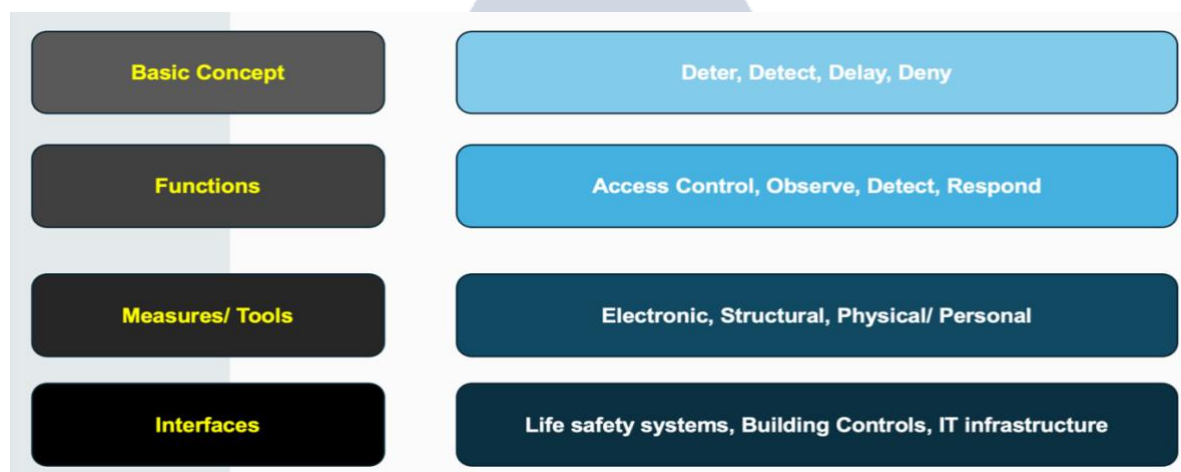
2.4. **Assessment** is the ability of the physical protection system to determine what was detected and make a decision about whether or not to initiate a response.

2.5. **Delay** is achieved through the deployment of security measures that slow the progress of an intruder.

2.6. **Response** refers to the actions taken to neutralize, contain, or mitigate an event.

2.7. **Evidence gathering** is essential to supporting the prosecution of intruders, and each prosecution is a good advertisement for the security of the site.

### 3. Relationship Between Physical Security Concepts, Functions & Measures.



## Chapter - 3 : Planning & Conducting Physical Security Assessments

### 1. General Risk Assessment Models & Considerations.



#### 1.1. Conditions that Tend to increase the exposure to Assets.

1.1.1. **Physical.** Includes factors such as the organization's types and locations of facilities or campuses, its surroundings.

1.1.2. **Nonphysical.** Includes factors such as the geo-political landscape; culture; industry pressures; legal, regulatory and compliance requirements; intensity of competition; organizational growth mode; speed of decision making; and willingness to adopt technology.

1.1.3. **Logical.** Includes information and digital assets and the network or digital space that connects them to each other and to their users and stakeholders.

1.2. This risk assessment process is meant to be a cyclical and continuous effort since the elements are constantly subject to change over time.

## 2. Qualitative & Quantitative Methods.

2.1. **Qualitative** analysis includes any approach that does not use numbers or numeric values to describe the risk components. Generally, comparative terms such as critical, high, medium, low, and negligible may be used to gauge the asset value and levels of risk components and risk itself. **This is most suitable when evaluating basic security applications.**

2.2. **Quantitative** analysis includes any approach that uses numeric measures to describe the value of assets or the level (severity or probability) of threats, vulnerabilities, impact, or loss events. It can vary from simple scale ratings (e.g., 1 to 5) to sophisticated statistical methods and mathematical formulas. This method is used to measure the effectiveness of a physical protection system whose primary functions are to detect, delay, and respond.

## 3. Assets.

3.1. Asset identification is critical to an estimate of risk at a site.



3.2. Asset valuation may be quantitative (numerical measurements), qualitative (interviews, surveys, and industry publications), or a mixture of both.

3.3. The impact of loss as measured in currency is best described as criticality.

3.4. Asset values can be quantified in different ways. One way to is assign them a relative value, such as a number from 1 (low) to 5 (high), based on priority. Another way is to apply a cost- of-loss formula:

$$K = C_p + C_t + C_r + C_i - I$$

K = total cost of loss

C<sub>p</sub> = cost of permanent replacement C<sub>t</sub> = cost of temporary substitute

C<sub>r</sub> = total related costs (remove old asset, install new, etc.)

C<sub>i</sub> = lost income cost. I = available insurance or indemnity

## 4. Evaluating Threats.

4.1. A threat is any verbal or physical conduct that conveys an intent or is reasonably perceived to convey an intent to cause physical harm or to place someone in fear of physical harm.

4.2. **Manmade threats.** Evaluation of intentional threats is based on identification and study of potential adversaries. Sources of information on adversary capabilities and intentions are past history, adversary sequence diagrams, intelligence and other open sources, internal communications (newsletters, Web sites, blog posts, etc.), law enforcement reports, crime analysis, professional organizations.

4.3. **Natural threats.** Natural threats are typically evaluated using historical trends and statistics. Comprehensive contingency planning and at least some degree of all-hazard preparedness is strongly recommended by most professionals.

## 5. **Vulnerabilities.**

5.1. Vulnerability is the state of being susceptible to **harm** or injury.

5.2. Vulnerabilities can be evaluated as under.

5.2.1. Observability is the ability of an adversary to see and identify a vulnerability.

5.2.2. Exploitability is the ability of the adversary to take advantage of the vulnerability once aware of it.

5.3. **In assessing natural threats, observability is reversed**, referring to security personnel's ability to observe, become aware of, or track an oncoming threat, such as a storm.

## 6. **Risk Analysis.**

6.1. The assessor combines all the information on assets, threats, and vulnerabilities and then considers the potential impact and prioritizes it based upon the consequences of a loss event.

6.2. **Risk = (Threat × Vulnerability × Impact)<sup>1/3</sup>**

6.3. Risk analysis results should be presented in a priority order.

## 7. **Risk Mitigation.**

7.1. The goals and mission of an organization should be considered in selecting a risk mitigation approach.

7.2. It is not practical to address all identified risks, so priority should be given to those risks that have the potential to cause significant mission impact or harm.

7.3. **Risk mitigation techniques: Avoid, Transfer, Mitigate/reduce, Accept**

## 8. **Physical Security Assessment.**

### 8.1. **Security Survey.**

8.1.1. A thorough physical examination of a facility and its systems and procedures, conducted to assess the current level of security, locate deficiencies, and gauge the degree of protection needed.

8.1.2. A comprehensive risk assessment focuses equally on assets, threats, vulnerabilities, and consequences, a **security survey places more emphasis on vulnerabilities**.

### 8.1.3. **Physical Security Assessment Methods.**

- **Outside-Inward Methodology** Under this approach, the assessment team takes on the role of an adversary (perpetrator) attempting to penetrate the physical defenses of a facility (penetration test). The team begins outer layer to inner critical asset.



- **Inside-Outward Methodology.** The assessment team takes on the role of the security professional (defender). They work from the asset or target out toward the outer perimeter.
- **Functional (Security Discipline) Methodology.** This approach addresses each security function or discipline individually and then collating the findings from the assessment components.

#### 9. **Assessment Results.**

- 9.1. Security reports should provide the risk owner with sufficient understanding of the risk and how it can influence the mission of the company.
- 9.2. The most in-depth and competent security survey may be useless if the results are not properly reported or applied.
- 9.3. The five criteria of good reporting are : **Accuracy, Clarity, Conciseness, Timeliness, Slat or Pitch.**

### **Chapter- 4 : Physical; Security Metrics**

#### 1. **Understanding Metrics.**

- 1.1. **Metric:** a measure based on a reference that involves at least two points, e.g, quantity over time. The primary goal of metrics is to facilitate insight into performance and operations.
- 1.2. Security metrics should tell us about the state or degree of safety relative to a reference point and what to do to avoid danger.
- 1.3. Metrics measure the **effectiveness and efficiency** of an organization's operations over time.

#### 2. **Designing a Metrics Program.**

- 2.1. The **Security MET** helps security professionals evaluate any metric according to the following criteria:
- 2.1.1. Operational : Cost, Timeliness, Manipulation.
- 2.1.2. Technical : Reliability, Validity, Generalizability.
- 2.1.3. Strategic : ROI, Organizational relevance, communication.
- 2.2. **Enterprise Performance Framework** : Effectiveness, Efficiency, Strategic Improvement.
- 2.3. **Metrics should be SMART : Specific, Measurable, Attainable, Repeatable, Time dependent.**

#### 3. **Physical Security Metrics.**

- 3.1. **Physical Security Systems Metrics.** Physical Access Control System (PACS) – Types of Alarms.
- 3.1.1. **Forced door.** A forced door is commonly defined as an entry through a controlled door **without authorization** or the **use of an authorized device**. This type of alarm can be initiated either by **user error, Malicious intent, system malfunction.**
- 3.1.2. **Door held open.** This type of alarm is generated when the predefined duration for the door position switch bypass expires. The PACS then considers the door to be held open or propped open. Other causes include **user error, system device malfunction.**
- 3.1.3. **Unauthorized access attempts.** These alarms may be the result of user error, system programming errors, or device malfunction. The information collected from these alarms can be used as a metric to diagnose system issues and also as an

investigative tool to identify patterns of access attempts at a specific location or by a specific individual.

3.1.4. Communications failure. This type of alarm is generated when one system component fails to communicate with another within a specified time. Causes : Network availability, Cabling/ wiring issues.

### 3.2. Physical Security Personnel Metrics.

#### 3.2.1. Response Metrics.

- **Response to an alarm or incident** by guard force or security system integrator response.
- **Response to customer requests.** How an office responds to its customers has a direct impact on the reputation.

#### 3.2.2. Training Metrics.

- **Inward-facing training metrics.** These cover training required for the security staff as part of their annual performance plan, position requirements, contract, or professional development.
- **Outward-facing training metrics.** Cover customer training requirements developed and delivered by the security office to the organization's employees or other audiences.

### 3.3. Physical Security Compliance Metrics.

- **Mean time to mitigate vulnerabilities (MTTMV).** This metric indicates the performance of the organization in reacting to vulnerabilities identified in the environment.

$$\text{MTTMV} = \frac{(\text{Date of mitigation} - \text{Date vulnerability was detected})}{\text{Total number of mitigated vulnerabilities}}$$

## Chapter - 5 : Basic Design Concepts

### 1. Design Principles.

- 1.1. The two foundational principles are the **four Ds (deter, detect, delay, and deny)** and **layered security** (defense-in-depth or security-in-depth).
- 1.2. A layer might be physical (such as a wall or entry control point) or functional.
- 1.3. The **critical detection point** is defined as the point where the delay time remaining exceeds the response force time.
- 1.4. Using a defense-in-depth approach means an adversary must avoid or defeat a number of protective devices or features in sequence.

### 2. Point Vs Area Security.

- 2.1. **Area security**, where there is significant perimeter protection and entry is tightly controlled at a single portal, or
- 2.2. **Point security**, where the outer perimeter is loosely controlled, and the primary security focus is at the individual buildings (or restricted areas within the buildings).

### 3. Design Criteria.

- 3.1. A design process must have criteria against which elements of the design will be evaluated
- 3.2. A **design process** based on performance criteria will select elements and procedures according to the contribution they make to overall system performance.

3.3. A **feature criteria** approach selects elements or procedures to satisfy requirements that certain items be present.

3.4. Performance measures.

**3.4.1. Detection**

- probability of detection
- time for communication and assessment
- frequency of nuisance alarms (false positives)
- frequency of undetected intrusions (false negatives)

3.4.2. **Delay** – Time to defeat obstacles.

**Chapter – 6 : Influencing Factors in Physical Security Design**

**1. Characteristic of the Assets Under Protection.**

1.1. an **individual** (such as a corporate executive, government official, or celebrity)

1.2. an item of **intellectual** property (such as a trade secret or physical model of a device)

1.3. an **event** or series of events

1.4. a vehicle (automobile, train, ship, aircraft, or other conveyance)

1.5. data stored or processed by a computer, IT network, device, or telecommunications system

1.6. a high-value object (eg., a painting, sculpture, jewel, cash, etc.)

animals, plants, bio-organisms, or other living things

1.7. Each asset type requires a different approach to physical security planning and implementation, as well as different metrics or criteria for measuring security effectiveness.

**2. Characteristics of the Building or Facility.** When the asset is a building or facility, a different set of influencing factors should be considered.

**2.1. Ownership and Occupancy.**

2.1.1. More leverage and flexibility are available when working directly with a building owner as opposed to a tenant.

2.1.2. an owner will take a greater interest in issues and details if the owner occupies the facility.

2.1.3. Special security arrangements may be needed for a building owner once the facility is occupied and operational.

2.1.4. In multi-tenant facilities, a more adaptable approach must be taken to physical security design, and this flexibility must continue throughout the life of the facility.

**2.2. Purpose of the Facility.**

2.2.1. How & by whom the facility will be used will influence the physical security strategy as well as the design & layout.

2.2.2. issues as access control, standoff distance, sensor types, nature of the security officer force, reliance on public utilities, and other factors depend on the purpose of the facility.

**3. Access.**

3.1. The intended access to a facility (private, public, or mixed) can have a significant influence on physical security strategies.

3.2. Features that can be affected are facility layout, parking arrangements, security systems, and the role or use of security officers.

3.3. **Hours of Business.** The hours of business will also define the security measures in terms of the access control measures to be implemented.

#### 4. Characteristics of Surroundings.

4.1. Security professionals must consider the facility's surroundings, including the terrain and neighbours.

4.2. Neighbours have a significant impact on the threats and vulnerabilities of a facility.

4.3. Neighbours may affect ingress and egress. A lockout or strike in neighbours facility may effect the ingress and egress.

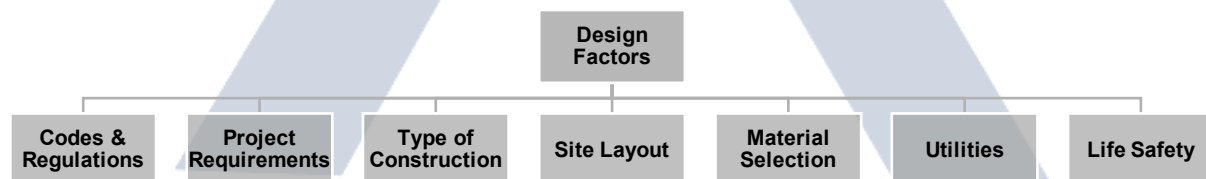
#### 5. Characteristics of Location.

5.1. Physical security may be implemented differently in a dense urban area than in a suburb, small town, or a rural or remote area.

5.2. The facility is near a body of water (this can be a positive or a negative factor), the reliability of supporting utilities and infrastructure, proximity to transportation facilities, accessibility byroads and bridges, telecommunications infrastructure, etc.

5.3. The propensity of the area for adverse weather and natural disasters is a key factor.

### Chapter – 7: Security Architecture & Engineering



#### 1. Codes & Regulations.

1.1. Projects are subject to a variety of federal, state, and local requirements.

1.2. Security designers must become aware of and follow all applicable standards and codes.

1.3. Series 4 of the United Facilities Criteria (UFC) pertains to security.

#### 2. Project Requirements.

2.1. A necessary part of developing a design is to conduct a risk assessment.

2.2. A simple equation for calculating risk is:

**Risk = Vulnerability & Threat x Asset Value**

2.3. **Design Basis Threat (DBT).** A profile of the type, composition, capabilities, and methods (tactics, techniques, and procedures), as well as the goals, intent, and motivation of an adversary, upon which the security engineering and operations of a facility are based.

2.4. Under the **FEMA model**, there are two basic classifications of threats or hazards: **natural and human-caused** (also known as intentional threats).

2.5. The probability of a natural event can be determined using **historical data**.

### 3. Type of Construction.

3.1. Requirements will be based on the use of facility.

3.2. **Type of construction** : Retrofit or renovation/ New construction.

3.3. The security designer should recognize the potential for the creation of failure points.

| <u>New Construction</u>                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Pros</u>                                                                                                                                                                                                                                                                     | <u>Cons</u>                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• Integrated system and facility</li> <li>• Based on own design criteria</li> <li>• Systems designed for and tested for compatibility</li> <li>• Warranted systems</li> <li>• Incorporate future expansion/growth into design</li> </ul> | <ul style="list-style-type: none"> <li>• Majority of costs are front-loaded</li> <li>• Requires full design of structure and systems</li> <li>• Must procure site</li> <li>• More time required to complete design phase and construction Permit requirements</li> </ul>        |
| <u>Renovation or Retrofit</u>                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                 |
| <u>Pros</u>                                                                                                                                                                                                                                                                     | <u>Cons</u>                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• Use existing structure</li> <li>• installation and minor construction (just the system and components)</li> <li>• Shorter project timeline typically</li> </ul>                                                                        | <ul style="list-style-type: none"> <li>• Intrusive to current activities</li> <li>• Capacity of existing utilities and systems may be insufficient</li> <li>• Cost may exceed new construction</li> <li>• Existing systems may be incompatible or difficult to match</li> </ul> |

### 4. Site Layout.

4.1. For perimeter security, the basic premise of design is the **four Ds: deter, detect, deny, delay.**

4.2. Standoff distance also increases the footprint of the site and may have environmental considerations in addition to the cost factors.

#### 4.3. **Perimeter.**

4.3.1. The first lines of defense is the perimeter of the property.

4.3.2. Depending on the level of protection required for the facility, perimeter protection may include a combination of a fence line, security officers, cameras, vehicle barriers, intrusion detection systems, signs, or natural barriers.

4.3.3. The most common perimeter defense is a fence.

4.3.4. A good practice when defining the perimeter of a property with a fence is not to place it right at the property line.

4.3.5. Landscaping should reduce in size as it gets closer to the facility,

#### 4.4. **Lighting.**

4.4.1. Certain minimum lighting requirements exist for each type of visual security used, from normal vision to specialized cameras such as infrared or digital systems.

4.4.2. **Metal halide lights best support colour CCTV** because it emits a white type of light, whereas high-pressure sodium lights give of a yellowish light.

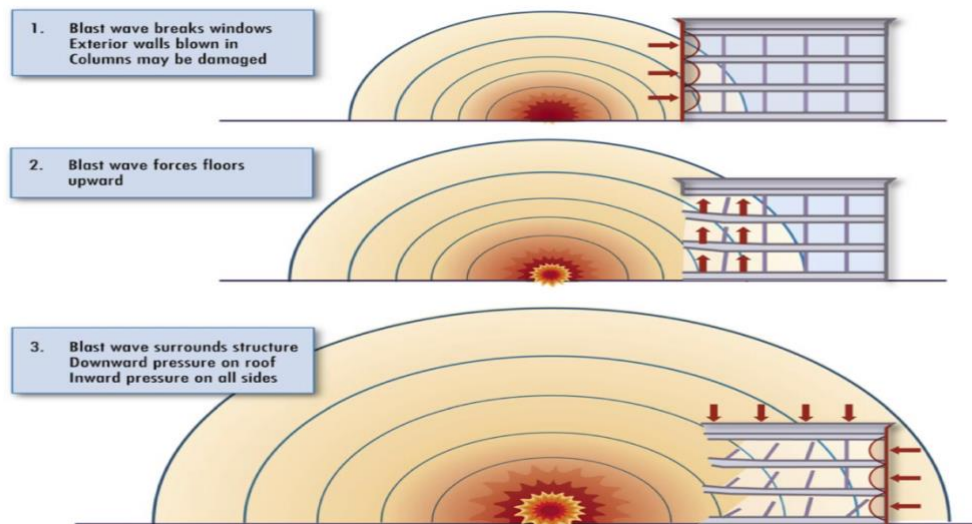
#### 4.5. **Building Design against Blast.**

4.5.1. These are some common terms related to pressure from a blast:

- **Incendiary thermal effect:** a bright flash or fireball at the instant of detonation.
- **Shock or blast wave:** the leading edge of the energy wave released by an explosion that rolls out in a circular pattern from the source.



- **Reflective pressure:** the energy of the shock wave that is redirected by objects along the initial path way; reflective pressures have the potential to be much stronger than the initial shock wave at the same distance from the detonation.
- **Overpressure:** the resulting amount of pressure above the ambient values.
- **Fragmentation:** materials encasing or near the explosives, thrown or projected by the pressure waves produced by the explosion.



**Blast Pressure Effects on a Structure FEMA, 2003**

#### 4.6. **Building Access.**

4.6.1. The building should have sufficient standoff, or distance from the threat, to ensure time for detection and security response.

4.6.2. Zones : **Perimeter, Non Exclusive Zone, Exclusive zone.**

4.6.3. When designing the site, it is useful to determine the time required to get a response from on-site security or local law enforcement.

#### 5. **Material Selection.**

5.1. The material used is important from an architectural standpoint, it provides the look of the overall design. From an engineering perspective, it provides strength to support the facility and protect the assets within.

##### 5.2. Building Structure.

5.2.1. All systems must be designed based on requirements established at the start of the project.

5.2.2. Structures are designed to protect assets against the design hazards.

5.2.3. **Progressive collapse** is the chain-reaction failure of a building's structural system or elements as a result of, and to an extent disproportionate to, initial localized damage, such as that caused by an explosion.

##### 5.3. Facades.

5.3.1. A facade is an important architectural feature that defines the look of the structure.

5.3.2. The materials selected should not create vulnerabilities or assist the attacker but should be neutral or act as a countermeasure.

#### 5.4. Interior Layout.

5.4.1. Interiors are designed with an open or closed (compartmentalized) floor plan or a combination of the two.

5.4.2. **Open floor** plans provide great visibility of all activities and personnel and leave few places for privacy. Ceiling height should be minimised.

5.4.3. **Closed or compartmentalized floor** plans provide privacy for personnel to conduct business. Awareness of activities occurring in the area is significantly reduced, and escape routes may be restricted.

5.4.4. Areas of special risk, such as mailrooms that process unknown hazards and threats, should be placed along exterior walls to reduce the potential for damage to the facility should an explosive detonate.

#### 6. Utilities.

6.1. Consideration for power redundancy, how utilities enter the site, maintenance needs, and access to the utilities needs to be incorporated.

6.2. If it is a critical utility, it makes sense to consider a design that loops or provides a redundant feed so if service is lost, it can be back-fed from another location.

6.3. Important to consider is how utilities are accessed by the service provider.

#### 6.4. **Emergency Power.**

6.4.1. During the design phase, it is important to determine what equipment or systems need to maintain power at all times and what length of interruptions can be tolerated.

6.4.2. Emergency power equipment requires regular maintenance, such as load tests on generators and inspections of UPS batteries.

#### 7. Life Safety Systems.

7.1. In the United States, life safety requirements are established through two different building codes. These are **the National Fire Protection Association (NFPA)**, specifically in its Life Safety Code (NFPA 101), and the IBC **International Building Code**.

7.2. **NFPA 101 provides requirements for various types of locking systems on doors.**

7.3. Emergency lighting is important to the successful evacuation of a facility.

7.4. **Evacuation.** The following are key steps to take when planning evacuations:

7.4.1. Designate primary and secondary evacuation routes.

7.4.2. Identify muster points away from the facility and other potential hazards.

7.4.3. Identify and train personnel-security personnel or employees-who will provide assistance during an evacuation.

7.4.4. Identify the locations of any special equipment that may be necessary during evacuation (e.g., escape ladders).

7.4.5. Identify the locations of fire extinguishers, smoke detectors, and first aid kits.

7.4.6. Identify the locations of the shutoffs for gas, water, and electricity.

**An evacuation route should be the easiest and closest unobstructed route to a nearby exit.**

**Pathways should be wide enough to allow safe passage of personnel without obstruction.**

## Chapter – 8 : Integrated Security & Protection Strategies

### 1. Integrated Electronic Security Systems.

1.1. Electronic security systems have video surveillance systems, access control systems, sensors, alarms, and other elements interacting with one another and often working in concert.

1.2. These security systems ride on a network backbone, which may be independent (the preferred approach) or part of an organization's enterprise network.

1.3. It is recommended that select a single source to take responsibility for the system's preventive and remedial maintenance.

### 2. Integrated Physical security Elements.

2.1. All elements of physical security must be integrated to operate as part of a comprehensive physical security strategy.

2.2. Physical security elements are generally categorized as **electronic security systems, structural security measures, and human capital.**

2.3. Security officers represent a key component of the strategy in most instances, but it is critical to match the right people with the specific role and requirements of the position.

### 3. Integrated Security Programs.

3.1. A successful security design recognizes technological developments on all fronts (not only electronic security systems) and integrates three primary elements: **architectural aspects, security systems, and operational factors.**

3.2. Technology does not replace manpower but rather acts as a force multiplier to augment personnel capabilities and provide checks and balances to offset individual wrongdoing.

3.3. The human element of security, which relates to decision-making, common sense, and awareness, must be integrated into the system during the **design phase.**

## Chapter – 9 : Structural Security Measures

### 1. Barriers.

1.1. A barrier is "a natural or man-made obstacle to the movement/direction of persons, animals, vehicles or materials".

#### 1.2. **Walls.**

1.2.1. Walls are generally more resistant to penetration than are doors, windows, vents, and other openings.

1.2.2. Reinforced concrete walls are commonly used in structures where sensitive materials are used or stored, and they are widely believed to be formidable barriers.

1.2.3. Placing two or more reinforced concrete walls in series results in longer penetration delays than using one wall that is as thick as the two walls combined.

1.2.4. Delay in penetrating a concrete wall can be increased by using additional rebar, increasing rebar size, or decreasing center-to-center rebar spacing.

#### 1.3. **Doors.**

1.3.1. Doors are the weakest form of barrier.

1.3.2. Most common exterior doors are 1 ¾ in. (44 mm) thick with 16 or 18-gauge (1.5 or 1.2 mm) steel surface sheets.

1.3.3. Steel pedestrian doors are found in single or double configurations.

1.3.4. Exterior doors swing outward and have their closing devices attached internally.

1.3.5. When complete door replacement is not an option, older standard doors can be upgraded.

1.3.6. At new facilities or where complete door replacement is necessary, new high-security, attack-resistant doors should be installed.

**1.3.7. *Eliminating unnecessary doors is the first step in upgrading a facility's resistance to penetration.***

1.3.8. Adding steel plates to the door surfaces increases penetration resistance against hand and light power tools.

1.3.9. Placing wood cores, especially redwood, between door plates increases the delay time for thermal cutting tools by three to four times that of an air void.

1.3.10. To prevent easy access to the lock/frame area, a sheet steel strip (**often called an astragal**) can be welded or bolted to the door.

1.3.11. Welding the pin top to the hinge extends penetration times if only hand tools are used.

1.3.12. Upgraded hinges with a stud-in-hole feature can extend penetration time.

1.3.13. To prevent hinge-side door removal is to bolt or weld a steel Z-strip to the rear face of the door.

#### 1.4. **Windows & Other Openings.**

1.4.1. Window glass typically used in commercial settings.

| <u>Type</u>                                 | <u>Composition</u>                                                                                                                             | <u>Examples</u>                                                    | <u>Breakage</u>                                        |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|
| Annealed or plate glass                     | manufactured to control residual stresses such that it can be subjected to fabrication                                                         | Regular plate, float, sheet, rolled, and patterned surface glasses | breaks into large shards that can cause serious injury |
| Tempered glass                              | treated to resist breakage.                                                                                                                    |                                                                    | Breaks fragments into small pieces rather than shards  |
| Wired glass                                 | resistance against large objects, but it may still shatter                                                                                     | Used to maintain fire ratings in <b>fire-rated doors</b>           |                                                        |
| Laminated glass                             | composed of two sheets of ordinary glass bonded to a middle layer or layers of plastic sheeting material                                       | preferred glass type for mitigating blast forces                   | Protects from glass shattering                         |
| Bullet-resistant or burglar-resistant glass | It is laminated and consists of multiple plies of glass, polycarbonate, and other plastic films to provide many levels of ballistic resistance |                                                                    |                                                        |

#### 1.4.2. **Measures to enhance resistance to Penetration.**

- For greater penetration resistance, thick security glass can be used.
- standard glazing materials are often upgraded with a protective grill of expanded steel mesh or other forms of metal grills.
- Window bars are steel bars that, where permitted by building and fire codes, can protect the window opening from being used as an access point.

- Window film (fragment retention film) adheres to the interior surface of the glass, strengthens and holds the glass in place if broken, and can accomplish various other purposes.
- Blast curtains are made of reinforced fabrics that provide protection from flying materials in an explosion.
- Security shutters can add to the protection of windows.
- The penetration resistance of windows and utility ports may be increased with grilles, bars, expanded-metal mesh, or screens.

### 1.5. Roofs & Floors.

1.5.1. Roofs and floors are constructed using similar methods, but they vary in thickness, type, and quantity of steel reinforcement, as well as in concrete strength required to carry the loads.

1.5.2. The following are among the methods of roof enhancement:

- enhancing membranes with embedded screen
- adding several inches of rigid insulation
- using concrete reinforced with deformed steel bars and expanded steel mesh
- forming larger rebar into several rows or layers for reinforced concrete
- increasing the number of fasteners and adding structural members to corrugated roofs
- using mechanical fasteners or joints and a continuous weld and heavier gauge material on metal roof systems
- using larger rebar to strengthen the flange area of precast concrete tee beams

1.5.3. Barriers placed below the roof may be more effective against penetration than those in the roof itself. **The optimum distance between the roof and the secondary barrier is 10 to 12 in.**

### 1.6. Fencing & Perimeter Walls.

1.6.1. The most common perimeter defense measures in both government and private sector applications is fences and perimeter walls.

#### 1.6.2. Chain Link Fencing.

Chain link fence fabric is made from wire wound and interwoven to provide a continuous mesh without knots or ties, except the knuckling or twisting and barbing the ends of the wires to form the selvage of the fabric.

Chain link fabric can be zinc coated, aluminium coated, or plastic/polymer coated.

ASTM Standards. Height – **4 Ft for preventing animals & 7 Ft for preventing humans. Signs placed every 50 Ft (15m).**

Chain Link Fence Installation.

| <u>Description</u> | <u>Salient Aspects</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Posts.</b>      | <p>Terminal posts are placed at the ends and corners of fences and to support gates.</p> <p><b>Pull posts</b> are terminal posts bracing a long stretch of fence or offering a change of elevation.</p> <p>All other posts are <b>line posts</b>.</p> <p>Line posts should be spaced at intervals not to exceed 10 ft. (3 m).</p> <p>The diameter of the post holes should be <b>four times</b> the largest cross section of the post.</p> <p>The depth of the post hole should be a minimum of 24 in. (0.6 m), plus an additional 3 in. (76 mm) for each 1 ft. (0.3 m) increase in fence height over 4 ft. (1.2 m).</p> |



|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bracing</b>             | <p>Terminal posts should be braced diagonally to the nearest line post when bracing is required.</p> <p><b>NO braces are required</b> on fabric 6 ft (1.83 m) high or less <b>when a top rail is installed</b>. Braces are required on all fabric over 6 ft. (2.83 m).</p> <p>On fabric over 12 ft. (3.66 m) in height, a centre rail is required.</p> <p>Regardless of height, any fence installed <b>without a top rail should have braces on all terminals</b>.</p>                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Fabric</b>              | <p>The fabric should be <b>No. 9 AWG</b> (American wire gauge) or heavier, and the <b>mesh openings should not be larger than 2 in. (50 mm)</b>.</p> <p>Fences 5 ft. high and under, selvage should be knuckled at both ends.</p> <p>Fences higher than 5 ft. selvage should be knuckled at one end and twisted at the other.</p> <p>Fences less than 2 in. mesh, all selvage should be knuckled.</p> <p>On soft ground, fabric should reach below the surface sufficiently to compensate for shifting soil.</p> <p>Breaking Loads.</p> <ul style="list-style-type: none"> <li>• No. 6 gauge (0.192 in. or 5 mm): 2,170 lb. or 984 kg</li> <li>• No. 9 gauge (0.148 in. or 3.8 mm): 1,290 lb. or 585 kg</li> <li>• No. 1 gauge (0.120 in. or 3 mm): 850 lb. or 386 kg</li> <li>• No. 1 gauge (0.113 in. or 2.9 mm): 750 lb. or 340 kg</li> </ul> |
| <b>Fence top treatment</b> | <p>A top rail is a horizontal member to which fabric is attached with ties or clips at intervals <b>not exceeding 2 ft. (60.9 cm)</b>.</p> <p>A top guard, which is an overhang of barbed wire along the top of the fence usually at a <b>45-degree angle</b>.</p> <p>Three strands of barbed wire are used, <b>spaced 6 in. (15.2 cm) apart</b>.</p> <p>The supporting barb arms should be of sufficient strength to withstand a <b>weight of 250 lb. (113 kg)</b></p>                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Gates</b>               | <p>The number of gates installed should be kept to the minimum consistent with efficient operation.</p> <p>All gates should be provided with locks.</p> <p><b>Single swing</b> gates for walkways.</p> <p><b>Double swing</b> gates for driveways.</p> <p><b>Multifold</b> gates for openings upto 60 Ft.</p> <p><b>Overhead</b> gates where room for swing gates is not available.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Turnstiles</b>          | <p>Turnstiles are designed for a variety of security and pedestrian control Applications.</p> <p>Two heights-waist high (about 36 in. or 0.9 m) and full height (about 7 ft. or 2.1 m).</p> <p>The full-height turnstile which completely surrounds individuals passing through, can be automated with a card access control system, keypad, or biometric device if desired</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### 1.7. Blocking Barriers.

1.7.1. Blocking barriers are most often designed to **control or prevent vehicles access** to a space.

1.7.2. **Passive barriers** are fixed in place, do not allow for vehicle entry, and are used to provide perimeter protection away from vehicle access points. **Eg: Fixed bollards, Jersey barriers, Trees, water obstacles, walls & fences.**

1.7.3. **Active barriers** are most often used at vehicle access control points within a perimeter barrier system or at the entry to specific buildings within a site to provide a barrier for vehicle screening or inspection. **Eg : Retractable bollards, rising/ rotating wedge systems, crash gates, crash beams.**

### 1.8. Bollards.

1.8.1. To reduce the visual impact, bollard height should typically ***not be more than 30 in. (0.76m).***

1.8.2. Bollard ***spacing should be between 36 and 48 in. (0.9 and 1.2 m).***

1.8.3. Bollards should be arranged in a linear fashion in which the centre of the bollards is parallel to the centre line of existing streets.

1.8.4. ***In no case should bollards exceed a height of 38 in. (1 m).***

1.8.5. Retractable bollards may be operated manually or by electric, hydraulic, pneumatic, or gas-assisted mechanisms.

#### 1.8.6. **ASTM Ratings.**

| <b><u>Vehicle Weight</u></b> | <b><u>Vehicle Speed</u></b> | <b><u>ASTM Rating</u></b> | <b><u>Dept of State Rating</u></b> |
|------------------------------|-----------------------------|---------------------------|------------------------------------|
| 6800Kgs/ 15000 lbs           | 30mph/ 48 kmph              | M 30                      | K 4                                |
|                              | 40mph/ 64 kmph              | M 40                      | K 8                                |
|                              | 50mph/ 80 kmph              | M 50                      | K 12                               |

### 1.9. Jersey Barriers.

1.9.1. It is a standardized precast concrete element originally developed in the 1940s and 1950s as a highway median barrier to prevent vehicle crossovers into oncoming traffic.

1.9.2. Jersey barriers are not easily adaptable.

1.9.3. Standard length : 12.5 & 20 ft.

## 2. **Container & Vaults.**

- A layer of physical security - often the one that is ***closest to the asset itself***- consists of ***containers, safes, and vaults.***

- The ***degree of protection*** from intrusion afforded by a vault or safe may be ***measured in terms of the lock resistance.***

### 2.1. **Safes.**

2.1.1. Underwriters Laboratories (UL) has a well-established protocol for testing and rating safes and protective containers.

2.1.2. ***UL standards require that a safe weighing less than 750lb. (340 kg) be anchored.***

### 2.1.3. **Record Safes for Fire Protection.**

- A record safe typically incorporates insulation that contains moisture to help dissipate a fire's heat so that the interior temperature does not rise to the point where records are destroyed.

- Underwriters Laboratories designates three classes of records protection equipment

- (containers) as follows:

- Class 350, specified for paper records and products
- Class 150, specified for magnetic media, tapes, discs, and photographic materials
- Class 125, specified for floppy discs and similar flexible computer media

- Class 350 container are generally more sensitive to humidity damage than are the magnetic, photographic, or disc media stored in Class 150 containers.

### 2.1.4. **Fire Resistant Safes.**

- Fire-resistant safes fall into one of three classes (350-4, 350-2, or 350-1), based on the time expectations for paper records to be protected in a fire.

- During the fire endurance test, the inside temperature of a safe is recorded and cannot exceed 350° F (177° C) at the specified time limit (four hours, two hours, or one hour).

- Additional test criteria include the ability of the container to provide the same temperature conditions after being dropped from a height of 30ft. (9.1 meters).
- Ability to avoid exploding when exposed to a sudden temperature of 2,000° F (1,093° C).

#### 2.1.5. Insulated Filing Devices.

- Insulated filing devices, designated Class 350-1 (formerly Class D) and Class 350-1/2 (formerly Class E) afford considerably less protection for records than the three levels of fire-resistant safes.
- Insulated filing devices are not designed for impact resistance and are **not drop tested**.

#### 2.1.6. Electronic Data Processing Record Protection.

- Magnetic media begin to deteriorate at 150° F (66° C).
- This type of container has been designed to protect electronic records against 125° F (52° C) or 150° F (66° C) and 80 percent humidity for four-, three-, two-, or one-hour periods.

### Summary of UL Designations & Labels for Fire resistant containers

#### **Fire-Resistant Safes for Paper Records**

350-4 (four hours)  
350-2 (two hours)  
350-1 (one hour)

#### **Insulated Records Containers for Electronic Media**

150-4 (four hours)  
150-3 (three hours)  
150-2 (two hours)  
150-1 (one hour)  
125-3 (three hours)  
125-2 (two hours)  
125-1 (one hour)

#### **Insulated Filing Devices**

Class 350-1 or ½ (one or half hour)  
Class 150-1 or ½ (one or half hour)  
Class 125-1 or ½ (one or half hour)

#### **Insulated File Drawers**

Class 350-1 (one hour)  
Class 150-1 (one hour)  
Class 125-1 (one hour)

#### 2.1.7. Burglary-Resistant Containers.

- The body is made of high-tensile, electrically welded steel with doors of high-tensile and armour plate steel designed to resist attacks using power-driven carbide drills, abrasive wheels, and other burglary tools.
- **1/2 Hour - UL class 350 Rate Safe:** The unit is heated for half hour to reach an exterior temperature of 1550 degrees.

- **1 Hour - UL class 350 Rate Safe:** The safe is heated for one hour to reach an exterior temperature of 1550 degrees, then put through the cool down test.
- **2 Hour - UL class 350 Rate Safe:** The safe is heated for two hours to reach an exterior temperature of 1550 degrees, then put through the cool down test.

## 2.2. Vaults.

- Vaults are specially constructed rooms or areas intended to limit access and provide protection to the assets in the space.
- The term vault also applies to specially constructed rooms or areas that are designed to protect the contents from fire and not necessarily theft.

### 2.2.1. Fire-Resistive Vaults

The principal U.S. standard for fire-resistive vaults is **National Fire Protection Association (NFPA) 232**.

Vault Construction standards.

**Reinforced concrete with steel rods** at least 0.5 in. (13 mm) in diameter, spaced 6 in. (15.2 cm) on centre and running at right angles in both directions.

- A structural steel frame protected with at least 4 in. (10.2 cm) of concrete, brickwork, or its equivalent tied with steel ties or wire mesh equivalent to No. 8 ASW gauge wire on an 8 in. (20.3 cm) pitch.

- Because fire resistance is determined by wall thickness, the minimum thickness of a **4- hour vault wall is 12 in. for brick and 8 in. for reinforced concrete.**

The minimum thickness for a **6-hour vault wall is 12 in. for brick and 10 in. (25.4 cm) for reinforced concrete.**

### 2.2.2. Media Storage and Protection.

- Magnetic media are often stored in the same vaults used for paper products.
- The media should be stored according to any manufacturer's specifications and/or industry standards.
- Digital media - such as memory sticks, USBs, CDs and DVDs — require a safe that can withstand 238 degrees F or more.
- Data such as computer backups, internal hard drives, and tapes are the most vulnerable; these require a safe that may withstand temperatures of 150 degrees F or more.

## 3. Locks & Locking Mechanisms.

### 3.1. Mechanical Locks.

3.1.1. A mechanical lock uses a barrier arrangement of physical parts to prevent the opening of a bolt or latch.

3.1.2. The functional assemblies of components are as follows:

- The bolt or latch that actually holds the movable part (door, window, etc.) to the immovable part (jamb, frame, etc.)
- The keeper or strike into which the bolt or latch fits.
- The tumbler array that constitutes the barrier or labyrinth that must be passed to move the bolt
- The key or unlocking device designed to pass the barrier and operate the bolt

### 3.1.3. Warded lock.

- The mechanical lock longest in use and first developed is the warded lock.
- The greatest weaknesses of this type of lock are its vulnerability to spring manipulation.

**3.1.4. Lever Lock.**

- The lever lock offers more security than the warded lock.
- Application today in such varying situations as desk, cabinet, and locker installations, bank safe deposit boxes, and U.S. mailboxes.
- Susceptible to picking.

**3.1.5. Pin Tumbler Lock.** Most widely used lock for exterior and interior building doors.

**3.1.6. Wafer Tumbler Lock.** the wafer tumbler lock uses flat tumblers fashioned of metal or other material to bind the plug to the shell. Their design permits master keying. In addition, wafer tumbler locks may be designed for double-bitted keys.

**3.1.7. Master Keying.** The principle of master keying is that a single lock may be operated by more than one key by designing various different keys to engage or work on different tumblers or different aspects of the same tumblers.

**3.1.8. Security Vulnerabilities of Mechanical Locks**

- **Attack by force.** This could mean separating the door or movable element from the jamb or immovable element.
- **Surreptitious attack.** This involves picking or manipulating the lock with various tools.
- **Attack by impression-making and try keys.** The idea is to produce a facsimile key.

**3.2. Electronic Locking Mechanisms.****3.2.1. Electric deadbolt.**

- The electric deadbolt is the oldest and simplest of all electrical locking devices.
- A solenoid (electromagnet) moves a deadbolt, typically mounted on a door frame, either into or out of a strike plate on a door.
- The mechanism can be either fail safe or fail secure.
- The electric deadbolt is not normally recommended for application to doors required to be unlocked automatically in response to a fire alarm signal.

**3.2.2. Electric latch.**

- Somewhat similar to an electric deadbolt is the electric latch.
- It has solenoid activated, mounts on the door frame, and uses a strike plate in the door.
- It has an advantage over the deadbolt because the latch does not need to be withdrawn for the door to close since it is pushed into the lock
- mechanism against spring pressure as it rides up and over the strike plate.

**3.2.3. Electric strike.**

- The electric strike operates as an adjunct to any standard mechanical lock.
- The operating principle is simple: electrical energy is delivered to a solenoid that either opens or closes a mechanical latch keeper or strike plate.

**3.2.4. Exit device.**

- Also known as a panic bar or crash bar, the exit device is commonly used on doors in the path of egress from structures with high occupancy.



- Exit devices can be electrified to permit remotely controlled re-entry via a push button or card reader/keypad.
- **Delayed egress locking system.** Developed as a compromise between safety and security, this system is usually applied to doors intended to be used only for emergency fire egress. Instead of allowing immediate egress when pushed, activation of the bar starts a 15 or 30 second delay, after which the door unlocks.

### 3.2.5. Electromagnetic lock.

- The electromagnetic lock, also known simply as a **magnetic lock or Maglock**, uses an electromagnet and a metal armature or strike plate.
- When energized, the magnet exerts an attractive force on the armature and thus holds the door closed.
- Magnetic locks are rated by the pounds of force required to separate the armature or strike plate from the electromagnet. Ratings range from **500 to 2,000 lb.**
- A valuable feature of regular electromagnetic locks is that they have no moving parts and are less maintenance-sensitive than mechanical or electromechanical devices.

## 4. Lighting & Security Applications.

### 4.1. Types of Lighting Equipment & Lamps.

4.1.1. **Streetlight.** This projects a downward, circular illumination.

4.1.2. **Searchlight.** This uses a very narrow, high-intensity beam of light to concentrate on a specific area.

4.1.3. **Floodlight.** This projects a medium to wide beam on a larger area.

4.1.4. **Fresnel.** This lighting typically projects a narrow, horizontal beam.

4.1.5. **High mast lighting.** This is used mainly in parking lots and along highways and usually varies from 70 to 150 ft. (21 to 46 m) in height and is the primary determinant of lighting levels outside the perimeter boundary.

4.2. The main lighting sources (lamp types) are as follows.

4.2.1. **Incandescent.** These lamps operate by passing an electric current through a tungsten wire that becomes white hot and produces light. **These are the least efficient and the most expensive to operate, and they have a short life span.**

4.2.2. **Halogen and quartz halogen.** These are incandescent bulbs filled with halogen gas. They provide about 25 percent better efficiency and life than ordinary incandescent bulbs.

4.2.3. **Fluorescent.** These lamps pass electricity through a gas enclosed in a glass tube to produce 40 to 80 lumens of light. They develop twice the light and less than half the heat of an incandescent bulb. Fluorescent lamps are not used extensively outdoors.

4.2.4. **Mercury vapor.** These lamps also produce light by passing a current through a gas. They take several minutes to produce full light output, and they have **poor colour rendition** for video surveillance because of the **bluish light they cast**, but they have a **long life**.

4.2.5. **Metal halide.** Another form of gaseous lamp, they are often used at sports stadiums because they imitate daylight conditions and colours appear natural. However, they are the most expensive lamp type to install and maintain.

4.2.6. **High-pressure sodium.** These lamps are energy efficient and have a long lifespan, but they suffer from poor colour rendition. They are often applied on streets

and parking lots, and their particular quality of light enables people to see more detail at greater distances in fog.

4.2.7. **Low-pressure sodium.** These lamps are even more efficient than high-pressure sodium, but they are expensive to maintain

4.2.8. **LED.** These lamps furnish a highly cost-effective alternative that lasts longer without sacrificing illumination.

4.2.9. **Induction.** Induction lamps have a long life and, like fluorescent lamps, are used mainly indoors, except for parking structures, underpasses, and tunnels.

#### 4.3. Characteristics of Light & Lighting.

4.3.1. Light level, intensity, or illuminance is the **concentration of light over a particular area.**

4.3.2. **Illuminance** is measured in lux (metric units) representing the number of lumens per square meter or foot-candles (fc).

4.3.3. One foot-candle is equal to **10.76 lux** (often approximated to a ratio of 1:10).

4.3.4. **Reflectance.** The measure of an object's reflectance is the ratio of the quantity of light (measured in lumens) falling on it to the light being reflected from it, **expressed as a percentage.**

4.3.5. **Corrected Colour temperature (CCT)** is a measure of the warmth or coolness of a light. It is measured in degrees Kelvin, which is the centigrade (Celsius) absolute temperature scale where 0° K is approximately -272° C.

| <u>Material</u> | <u>Reflectance</u> |
|-----------------|--------------------|
| Asphalt         | 5%                 |
| Concrete (old)  | 40%                |
| Concrete (new)  | 25%                |
| Red brick       | 25%                |
| Grass           | 40%                |
| Snow            | 95%                |

4.3.6. **Colour Rendition Index (CRI).** The ability of a lamp to discriminate, grade, and faithfully reproduce the colours seen in an object is known as colour rendition and is measured as a CRI on a scale of 0 to 100. A CRI of 70 to 80 is considered good, above 80 is considered excellent, and 100 percent is considered daylight.

| <u>Lamp Type</u>     | <u>CRI</u> |
|----------------------|------------|
| Incandescent         | 100        |
| Halogen              | 100        |
| Fluorescent          | 75- 100    |
| Metal Halide         | 70         |
| LED                  | 75- 90     |
| Mercury Vapor        | 50         |
| High Pressure Sodium | 20         |
| Low Pressure Sodium  | 05         |

#### 4.4. Overview of Lighting System.

4.4.1. **Lamp (also known as a light bulb).** This is the manufactured light source, which includes the filament or an arc tube, its glass casing, and its electrical connectors.

4.4.2. **Luminaire (also Known as fixture).** This is the complete lighting unit, consisting of the lamp, its holder, and the reflectors and diffusers used to distribute and focus the light.

4.4.3. **Mounting hardware.** Examples would be a wall bracket or a light pole used to fix the luminaire at the correct height and location.

4.4.4. **Electrical Power.** This is what operates the lamp, ballasts and photocells.

#### 4.5. **Starting & Restrike.**

4.5.1. An extended relighting time is typical of high-intensity discharge (HID) lamps since they rely on an arc to produce light.

| <b>Lamp Type</b>                        | <b>Start Time (Mins)</b> | <b>Restrike Time (Mins)</b> |
|-----------------------------------------|--------------------------|-----------------------------|
| Incandescent, halogen, Fluorescent, LED | Instant                  | Instant                     |
| Metal Halide, Mercury Vapour            | 5-8                      | 10-20                       |
| High Pressure Sodium                    | 2-5                      | 1-20                        |
| Low Pressure Sodium                     | 5-8                      | 0-8                         |

#### 4.6. **Security Lighting for selected Applications.**

4.6.1. The four applications for security lighting are **standby, continuous, movable, and emergency.**

4.6.2. For pedestrians or normal cameras, the minimum level of reflected light for **detection is 0.5fc, For recognition is 1.0 fc, identification is 2.0fc.**

4.6.3. **Perimeter Fencing.** NRC regulation specify 0.2fc of illumination.

4.6.4. **Site Landscape & approaches.** Typical lighting levels for walkways are **1-4 fc for walkways, 0.5-2 fc for roadways, 10 fc for entrances, and 2 fc for open yards.**

4.6.5. **Building facade.** Typical lighting levels for security are **0.5-2 fc.**

4.6.6. **Parking structures.** A horizontal illuminance level of 5 fc with a uniformity ratio of 4:1 provides an adequate level of security.

4.6.7. **Open Parking.** Recommended light levels range from a minimum of 0.2 fc in low-activity general parking and pedestrian areas to 2 fc in high-activity vehicle areas. Cash collection and vehicular access control areas should be maintained at a minimum of 5 fc.

4.6.8. **Security control and monitoring rooms.** 30-50 fc with task areas at 50 to 70 fc.

4.6.9. **Guard and gate houses.** The area stounding a gate or guard house should be well lit, 2-5 fc, on the exterior at night.

### **Chapter -10 : Crime Prevention Through Environmental Design (CPTED)**

#### 1. **Principles of CPTED.**

1.1. CPTED's central principle is that the proper design and effective use of the built environment can lead to a reduction in the incidence and fear of crime, as well as an improvement in the quality of life.

1.2. CPTED is based on common sense and a heightened awareness about how people use their space for legitimate and criminal intentions.

1.3. CPTED is a set of management tools that target the following:

1.3.1. **Places.** Physical environments can be designed to produce behavioural effects that reduce the opportunity for certain types of crime and the fear of those crimes.

1.3.2. **Behaviour.** Some locations seems to predict, create, promote, or allow criminal activity or unruly behaviour while other environments elicit compliant and law-abiding conduct.

1.3.3. **Design and use of space.** Redesigning a space or using it more effectively can encourage desirable behaviour and discourage crime and related undesirable conduct.

1.4. The three underlying elements of CPTED are territoriality, surveillance, and access control.

1.5. CPTED measures are :

1.5.1. **Mechanical**-involving physical security hardware or electronic systems such as locks, fencing, or CCTV.

1.5.2. **Organizational**-involving people or activities rather than equipment, such as security guards, police patrols, or neighbourhood watch programs

1.5.3. **Natural**-involving natural features such as terrain, layout, landscaping, gates, walls, and other nonmechanical objects.

1.6. CPTED tools:

1.6.1. **Natural territorial reinforcement.** This is the process of establishing a sense of ownership, responsibility, and accountability in property owners, managers, or occupants to increase vigilance in identifying trespassers. Territorial reinforcement measures, which may be physical or symbolic, tell people they are in a defined space.

1.6.2. **Natural surveillance.** Increasing visibility by occupants and casual observers increases the detection of trespassers or misconduct at a facility. Windows, door viewers, mirrors, and other design feature that improve visibility fall under natural surveillance.

1.6.3. **Natural access control.** The idea is to employ both real and symbolic barriers-including doors, fences, and shrubbery-to define and limit access to a building or other space.

1.6.4. **Management and maintenance.** For spaces to look well cared for and crime-free, they must be maintained. The "**broken windows**" theory suggests that leaving broken windows or other decay markers unattended or unrepaired can lead to the impression of abandonment and increase crime opportunity because no capable guardian is observed.

1.6.5. **Legitimate activity support.** Some places are difficult to protect by nature of their location or other geographic features. In such instances, legitimate activity support is essential.

## 2. **Criminal Behaviour & Patterns.**

2.1. Applying CPTED requires an understanding of basic crime prevention theory and practice.

2.2. The likelihood of crime increases when a potential criminal feels the chances of detection and identification are low and the chances of escape are high.

### 2.3. **Steps that can assist in preventing repeat victimization:**

2.3.1. **Quickly remove signs of victimization.** It is important to remove or repair obvious signs of property damage as quickly as possible.

2.3.2. **Improve physical security.** Doing so enhances natural surveillance, visibility, sightlines and access control.

2.3.3. **Block easy access to targets.** This can be done by installing doors, gates, screens, and other real or symbolic barriers to make the targets more difficult to access and any valuables more difficult to remove.

2.3.4. **Protect especially vulnerable targets.** Some targets cannot be moved, but removable bollards, roll-down gates, fences, and screens can deny access to them after hours.

2.3.5. **Regulate access to high-risk assets or areas.** It may be necessary to require identification cards, permits, or fees for access to areas like bathrooms, parks, schools, and parking garages.

2.4. **Capable Guardian Concept.** Routine activity theory suggests that the presence of capable guardians may deter crime.

### 3. **Second Generation CPTED.**

3.1. Second generation CPTED refocused attention to encompass not only reducing physical opportunities for crime but also creating a sense of neighbourliness.

#### 3.2. **Second generation CPTED employs four main strategies;**

3.2.1. **Cohesion.** Techniques include community groups, neighbourhood associations, and personal development programs.

3.2.2. **Capacity threshold.** Also known as upping point theory, this strategy balances land uses and urban features.

3.2.3. **Community culture.** Cultural, artistic, sporting, and other recreational activities bring neighbourhood people together in common purpose.

3.2.4. **Connectivity.** Strategies link the neighbourhood to surrounding neighbourhoods and to funding and political support from corporations and upper levels of government.

4. **Third Generation CPTED.** Modern-day CPTED is focused on introducing environmental sustainability and green technology to CPTED practice.

5. **CPTED 3-D and Beyond.** A concept that looks at CPTED through the lens of designation of a space, its definition in terms of management and identity, and design as it relates to desired function and behaviour. It incorporates the more traditional "4 Ds" (deter, detect, delay, and deny), as well as some other key considerations.

### 6. **Tools of CPTED.**

#### 6.1. **Territoriality.**

6.1.1. This concept contends that people take better care of space when they feel some degree of ownership or responsibility over it-whether formal or informal.

6.1.2. The most basic tool of territoriality is an effective training and awareness program.

6.1.3. Another relatively simple tool is signage. Signage and graphics can represent very effective means of communication.

6.1.4. A **graphic** is a symbol that conveys a message pictorially.

6.1.5. **Signage** refers to conveying a message with letters or words.

6.1.6. Considerations affecting the use of security signage and graphics:

- **Architectural design considerations.** A sign to be clearly read by a person with 20/20 vision at 50 ft. (15 m), the letters must be at least 6 in. (15 cm) high, and graphics or symbols must be at least 15 in. (38 cm) high.

- **Systems considerations.** Graphics and signage should be consistent, uniform, and well distributed.

6.2. **Surveillance.** Surveillance is the act of observing. Surveillance can also be natural, a form that can be just as effective in meeting certain protection objectives as electronic surveillance systems.



6.3. **Access Control.** Access control can be natural. One of the oldest examples of security design (and also CPTED) is a **castle**.

6.3.1. The following are several types of security zones:

- **Unrestricted zones.** Some areas of a facility should be completely unrestricted to persons entering during the hours of designated use. Unrestricted zones might include lobbies, reception areas, snack bars, certain personnel and administrative offices, and public meeting rooms.
- **Controlled zones.** In these zones, a person must have a valid purpose for entry. Once admitted, the person may travel from one department to another without severe restriction.
- **Restricted zones.** These are sensitive areas limited to staff assigned to those areas. Sections within restricted zones may require additional access control.

## Chapter – 11 : Electronic Security Systems

### 1. Access Control Systems.

1.1. **Electronic access control systems** validate one or more credentials, which can be in the form of **something you know, something that is inherent to you, or something you carry**.

1.2. **Personnel access control** verifies based on determining whether the person is **carrying a valid credential, knows a valid personal identification number, possesses the proper unique physical characteristic that matches the person's characteristic recorded at enrolment**.

1.3. Means of Defeating Access Control Points :

1.3.1. **Deceit.** The adversary employs false pretenses in an attempt to convince security

1.3.2. **Direct physical attack.** Force and tools may be used to enter an area.

1.3.3. **Technical Attack** : Forged credential.

1.4. **Personal Identification Number.**

1.4.1. A four or six digit number is commonly used.

1.4.2. There are two primary considerations for selecting a secure PIN: it should be long enough, and it should not be a number that is particularly meaningful to the individual to whom it is assigned.

1.5. **Credentials.**

1.5.1. **Photo Identification Badge.** The photo identification badge is a common credential used for personnel access control, but it is not always effective. This kind of badge is manually checked, guard inattentiveness can reduce its effectiveness, especially at times when large numbers of people are entering a facility.

1.5.2. **Exchange badge.** A badge exchange system requires that matching badges be held at each access control point. When an employee presents a badge and requests entry, a guard compares the individual to the photo on the corresponding exchange badge held at the access control point. If the two match, the guard exchanges the badges and allows entry.

1.5.3. **Stored-image badge.** The use of a stored-image (video comparator) system requires a guard to verify an individual's identity based on visual characteristics. A securely stored image is used for comparison with a real-time image of the individual requesting entry.

#### 1.5.4. Coded Credential.

- **Magnetic stripe encoding.**

- This is widely used in credit and debit card systems.
- A strip of magnetic material located along one edge of the badge is encoded with data.
- The measure of the resistance of a magnetic material to changes in the stored information when exposed to magnetic field is called its **coercivity**.
- **Coercivity of Credit cards is 300 oersted (Low), High is 2500 – 4000 oersted.**
- Credential forgery is relatively easy since data from the magnetic stripe can be decoded.

- **Wiegand wire technology.** The Wiegand signal output format has become a de facto industry standard after it first emerged in the 1980s. While this technology is not used much anymore, the Wiegand data protocol is still in common use.

- **Open Supervised Device Protocol (OSDP).** OSDP is an access control communication protocol that improves interoperability among access control and security products. It became an international IEC standard and was adopted by the Security Industry Association (SIA) in 2020.

- **Bar code.**

- The bar code, widely used in retail trade to identify products at the point of sale, is sometimes used on coded credentials.
- The varying widths of the bars and spaces between them establish the code.
- Unless the bar code is covered with an opaque covering, it is relatively easy to duplicate.
- 2-D bar codes are also used on security credentials and are capable of storing more information than their one-dimensional counterparts.

- **Proximity Badge.**

- This is one whose information can be read without the badge being physically placed on to the reader device.
- Classified based on : **Power, operating frequency, read only / read or write capability.**
- Proximity badges fall into two groups according to frequency: **Low-frequency : 125 kHz and high-frequency from 2.5 MHz to over 1 GHz.**

- **Smart card.** This is the size of a standard credit card with an integrated circuit embedded in the card. There are contact and contactless smart cards. Contactless smart cards use RF to communicate with the reader and do not have metallic contacts.

## 2. Biometric Locks.

2.1. Biometric locks function by verifying a person's specific physical characteristic, such as fingerprint, hand geometry, face, iris, and retina.

2.2. Biometric devices can differentiate between **verification and recognition**. In **verification** mode, a person initiates a claim of identity, presents the specific biometric feature for authorization, and the equipment agrees. In **recognition** mode, the person does not initiate the claim; the biometric device attempts to identify the person, and if the biometric information agrees with the database, entry is allowed.

### 2.3. Types of Errors.

2.3.1. **Type - 1:** Type 1 error, also called a **false reject**, is the improper rejection of a valid user.

2.3.2. **Type - 2 :** A false accept, improper acceptance of an unauthorised person.

2.3.3. **Crossover point** where Type I errors equal Type II errors.

### 2.4. Types of Biometric Technologies.

2.4.1. **Eye pattern.** One technology uses the iris to accomplish identification. **2%** persons cannot be enrolled due to blindness or iris damage.

2.4.2. **Hand or finger geometry.** Personnel identity verification using the hand or finger geometry system is based on characterizing the shape of the hand. The underlying technique measures three-dimensional features of the hand, such as the widths and lengths of fingers and the thickness of the hand. During verification, the feature vector is compared with previous measurements.

2.4.3. **Fingerprints.** Fingerprints considered one of the most reliable means of distinguishing one individual from another. The systems rely on image processing and pattern recognition, have application in personnel access control. Optical methods using a prism and a solid-state camera are most often used to capture the fingerprint image.

2.4.4. **Voice.** Speech measurements useful for speaker discrimination include waveform envelope, voice pitch period, relative amplitude spectrum, and resonant frequencies of the vocal tract.

2.4.5. **Face.** Facial verification systems use distinguishing characteristics of the face to verify a person's identity. Most systems capture the image of the face using a video camera, although one system captures a thermal image using an infrared imager. Distinguishing features are extracted from the image and compared with previously stored features.

## 3. Contraband Detection.

### 3.1. Metal Detectors.

3.1.1. **Magnetometer.** The magnetometer is a passive device that monitors the earth's magnetic field and detects changes to that field caused by the presence of ferromagnetic materials. Materials such as copper, aluminium, and zinc are not detected.

3.1.2. **Pulse Field Detectors.** Continuous-wave detectors generate a steady-state magnetic field within the frequency band of **100 Hz to 25 kHz**. Pulsed-field detectors generate fixed frequency pulses in the **400 to 500 pulse-per-second range**.

3.2. **Package Search.** A conventional single-energy-transmission X-ray package search system produces an image for an operator to inspect. This is effective when the operator is effectively trained.

### 3.3. Explosive Detection.

#### 3.3.1. Bulk Explosive Detection.

- **Bulk explosives detection** technologies measure characteristics of bulk materials, thereby screening for the presence of explosives. The bulk characteristics that may be measured are the X-ray absorption coefficient, the X-ray backscatter coefficient, the dielectric constant and microwave or infrared emissions.

- **Backscatter technology** can image low-Z using the relatively large amount of X-ray energy scattered back in the direction of the source by low Z materials.

- **Computed tomography (CT)** is an automated technology detects small threat masses. The X-ray source and detectors are mounted on a gantry that spins around

the package, imaging the contents from different angles. CT scanners are the only X-ray approach that can extract enough information to calculate the material's mass, density, and mass absorption coefficient. **CT also suffers from relatively high nuisance alarm rates (up to 20 percent).**

- For vehicle and cargo-container searches, high-energy X-ray devices are available. The high-energy illumination is highly penetrating, allowing a reasonable image to be produced through the engine compartment or the filled trailer of a commercial truck.
- **Low-dose backscatter X-ray devices** can safely examine people for hidden items, providing an image of the body beneath the clothes. **The radiation dose to a person being screened is about 10 microrem.**
- **Quadrupole resonance (QR)** is a promising commercial technology that uses pulsed low-energy radio waves to determine the presence of nitrogen-rich materials. QR is very sensitive (detects small threat masses) for some explosives.

### 3.3.2. Trace Explosive Detection.

- **Ion mobility spectrometer (IMS).** In an IMS, the molecules in an air sample are negatively ionized using a radioactive Ni-63 source and chloride dopant, then passed into a drift cell through a shutter, which opens periodically (about every 20 milliseconds). While common high explosives form negative ions, some of the emerging explosives threats like triacetone-triperoxide (or TATP) also form positive ions. IMS-based detectors provide high sensitivity (nanogram quantities) to dynamite, military-grade TNT.
- **Chemiluminescence detector.** This technique uses photochemical detection. The vapor sample is collected and separated into its components using a fast gas chromatograph. The sample is then heated so that any nitrogen compounds present will decompose to form nitrogen oxide (NO). Chemiluminescence detectors have excellent sensitivity (picogram quantities) to common high explosives, including compounds with very low vapor pressures such as RDX and PETN.
- **Mass spectrometry.** In mass spectrometry, ions are processed in magnetic and electric fields to determine their mass-to-charge ratio. Quadrupole mass spectrometry and quadrupole ion trap time of flight are two examples of this method.

## 4. Physical Intrusion Detection System.

4.1. **Performance Characteristics.** Three main characteristics of intrusion sensor performance are probability of detection (PD), nuisance alarm rate, and vulnerability to defeat.

### 4.1.1. Probability of Detection.

- A perfect probability of detection would be 1.
- The two values PD and confidence level (CL) are used to describe the effectiveness of the sensor.
- The probability of detection depends primarily on these factors:
  - Target to be detected
  - Sensor hardware design
  - Installation conditions
  - Sensitivity adjustment
  - Weather conditions
  - Condition of the equipment

### 4.1.2. Nuisance Alarm Rate (NAR).

- A nuisance alarm is any alarm not caused by an intrusion.

- In an ideal system, the nuisance alarm rate would be zero.
- Causes of Nuisance Alarms: Vegetation, wildlife, weather conditions.
- ***In designing a system it is important to specify the acceptable False Alarm Rate (FAR).***

#### 4.1.3. Vulnerability to Defeat.

- Ideal sensor should not be defeated.
- Ways to defeat a system : ***Bypass, Adversary path exploitation, Spoofing.***

#### 4.1.4. Alarm Initiation Conditions.

- Conditions : ***Intrusion, Change in safety or process, Loss of power, Tampering/ Faults in system, Failure of the sensor.***

### 4.2. Standards.

4.2.1. **Underwriters Laboratories (UL) Standards.** Prepares safety standards as a guide to device manufacturers.

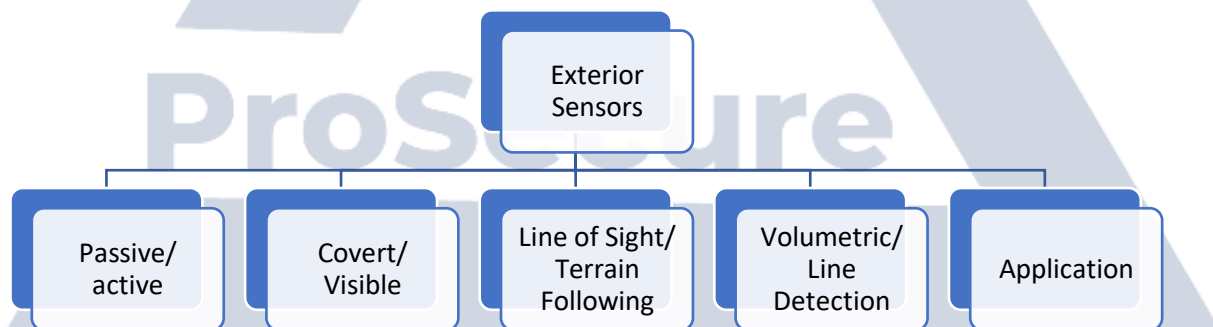
4.2.2. **American Society for Testing and Materials (ASTM).** Has established a committee to deal with security standards (Committee F 12). ASTM standards have not yet been developed for security alarm systems or sensors.

4.2.3. **European Alarm Standards.** Four intruder alarm grades.

| Grade   | Detail                                             | Alarm Suitability   |
|---------|----------------------------------------------------|---------------------|
| Grade 1 | Intruders have little knowledge and limited tools. | Low risk            |
| Grade 2 | Intruders have little knowledge and some tools.    | Low to medium risk  |
| Grade 3 | Intruders have knowledge and full range of tools.  | Medium to high risk |
| Grade 4 | Intruders have sophisticated knowledge and tools.  | High risk           |

## 5. Exterior Sensors.

### Classification of Exterior Sensors.



Process | Performance | Protection

### 5.1. Passive or Active.

#### 5.1.1. Passive Sensors.

- Operate in two manners - detect energy emitted by the object of interest or detect a target caused change in the natural field of energy.
- Use a receiver to collect energy emissions.
- Types of sensors – Vibration, heat, sound or capacitance.

#### 5.1.2. Active Sensors.

- The energy is transmitted and the changes in the received energy are detected.
- Contain both a transmitter and a receiver.



- Types of sensors – Microwave, infrared & Radio Frequency (RF).
- Fewer nuisance alarms as compared to passive devices.

## 5.2. **Line of Sight or Terrain Following.**

5.2.1. **Line of Sight.** The sensor requires clear line of sight between the transmitter and receiver.

5.2.2. **Terrain Following.** The radiated field follows the terrain and facilitates uniform detection.

## 5.3. **Volumetric or Line Detection.**

5.3.1. **Volumetric.** Generates alarm when an intruder enters the detection volume.

5.3.2. **Line Detection.** Detects motion along a line. Easy for an intruder to identify.

5.4. **Application Sensors:** Grouped by mode of application: Buried line (in the ground), Fence-associated (mounted on a fence or forms a sensor fence), Freestanding (neither buried nor associated with a fence but mounted on a support in free space)

### 5.4.1. **Buried Line Sensors**

**Pressure or seismic:** Covert, passive, terrain-following buried in the ground. Pressure type uses a reinforced hose filled with a pressurized liquid and seismic uses a string of geophones (they consist of a conducting coil and a permanent magnet). Both types lose their sensitivity in frozen soil.

5.4.2. **Magnetic field sensors:** Covert, passive, terrain-following buried in the ground. Wire loops buried in the ground sense change in a magnetic field and are effective in detecting vehicles or weapons.

5.4.3. **Ported coaxial cable sensor:** Covert, active, terrain-following. Detect human body and vehicles. Work well in frozen ground.

5.4.4. **Fiber-optic sensor:** Woven in a grid and buried just below the soil

## 5.5. **Fence-associated sensors**

5.5.1. **Fence-disturbance:** Visible, passive, and terrain-following to be installed on a security fence typically constructed with chain-link mesh

5.5.2. **Sensor fences:** Visible, passive, and terrain following Typically taut wire and less NAR than fence disturbance.

5.5.3. **E-Field or capacitance sensors:** Detect changes in capacitive coupling. Sensitive to lighting, rain, and small animals. Difficult to dig under/bridge overdue to the volume extension. Freestanding

5.5.4. **Infrared:** Active IR requires multiple beams to be effective (PD is then very high). Fog, dust, snow, etc. can cause NAR to increase. Passive IR also available

5.5.5. **Microwave:** Bistatic or monostatic is LOS and detect humans crawling. Standing water can cause a nuisance alarm

5.5.6. **Video Motion:** Many NAR issues

5.5.7. **Dual-Tech:** PIR & mono-static microwave

## 6. **Video Surveillance.**

6.1. Video surveillance has both primary and secondary use cases, making them a dual-purpose solution. Primary use cases support physical security, while secondary uses include business applications.

6.2. The main elements of a video surveillance system are as follows:

6.2.1. **Field of view.** This is the area visible through the camera lens.

6.2.2. **Scene.** This is the location or activity to be observed.

6.2.3. **Lens.** The lens determines the clarity and size of the field of view.

6.2.4. **Camera.** The camera converts the optical image provided by the lens to an electronic signal for transmission.

6.2.5. **Transmission medium.** This is the medium through which the signal generated from the camera is transmitted to equipment for viewing or recording, typically over coaxial cable, twisted-pair wire, network cable, optical fiber, or radio frequency signal.

6.2.6. **Workstation.** The client machine or workstation incorporates a monitor, computer, and keyboard/mouse or other control equipment. Workstations can be either fixed or mobile.

6.2.7. **Recording equipment.** This includes recorders and equipment for selecting which images to record, the speed at which the images will be recorded, the resolution of the capture, and the compression format for the capture.

6.3. **Functional Requirements.**

6.3.1. Camera Functional requirements to consider the following factors – Target & Activity.

6.3.2. Operational Requirement – Identification or observation. The final size of a video image is determined by the lens, the focal length, and image sensor.

6.4. One should base the choice of camera based on:

6.4.1. **Sensitivity:** the minimum amount of visible light necessary to produce a quality image.

6.4.2. **Resolution:** defines the image quality from a detail or reproduction perspective.

6.4.3. **Features:** the aspects that give one camera an advantage over another.

6.5. **Uses of VSS in Security.**

6.5.1. **Subject Identification.** This is the ability to identify something or someone within the scene beyond a shadow of a doubt. Subject identification depends first on the size and detail of an image and second on the angle of view. For subject identification, the object should occupy at least 10 percent of the scene's width.

6.5.2. **Action Identification.** This form of identification captures what happened.

6.5.3. **Scene Identification.** Each scene should stand on its own merit and is important for the security professional to witness and initiate appropriate action.

6.6. **Analog System Components.**

6.6.1. An Analog video system consists of three main components:

- **Camera** - used to transform a reflected light image into an electronic signal.
- **Transmission cable** - used to transmit the electronic video signal from the camera to the monitor.
- **DVR** (digital video recorder) is an electronic device that records video in digital format to a hard drive or disc drive. Record the number of cameras and the same time use search playback of recorded Video monitor (used to translate the electronic video signal into an image on a screen).

6.6.2. **Other Parts of the Analog video system.**

- **Pan or pan/tilt unit**
- A pan/tilt unit moves the camera from side to side and up and down.

- Pan/tilts can be an important part of a CCTV system, but in most applications, it is more cost-effective to use several fixed cameras instead of a single pan/tilt camera.
- **Controller**
  - A controller commands a function of a pan unit, pan/tilt unit, or automatic lens. In selecting a controller, it is important to remember that the application chooses the equipment, not the other way around.
- **Switcher**
  - To show the displays from several cameras on one or more monitors, a system generally requires a switcher.
  - Video switchers save money by making it possible to use more cameras than monitors.
- **Lens**
  - If it focuses light onto a chip or tube within a camera, it is a lens. Lenses come in a variety of sizes, allowing many different fields of view (the width and height of the scene). The proper lens makes it possible to capture an image that provides the right amount of identification within the overall scene.
- **Video transmitter/receiver**
  - This type of device allows the video signal to be transmitted via cable, phone line, radio waves, light waves, or other means. It is common to use several video transmission methods within a single system.
- **Coaxial cable** is the most common medium but not necessarily the best for all cameras within a system.
- **Amplifier**
  - Amplifiers strengthen the video signal for long-distance runs.
- **Video recorder**
  - This device retains the video information on a magnetic tape, CD, DVD, hard drive, or another medium. The right form of recording is determined by the need. Recording features include time-lapse, event-triggered, 24-hour, 72-hour, and more.

#### 6.6.3. System Design.

- **Keep the system in perspective.** CCTV systems contain only three major components: camera, cable, and monitor. Any other item is peripheral.
- **Let the application choose the equipment, not the other way around.** The user should first determine his or her needs and only then select equipment.
- **Design generically.** The best system design does not necessarily specify models or brands but remains open based on-site or scene requirements.
- **Design for the best option first.** Anything can be done with the right resources. It is best to design the application before establishing a budget. After the design is completed, the designer should calculate its cost and then, if necessary, remove some elements.
- **Don't feel the system must be built all at once.** Once the design is completed, the installation may be stretched over several budget years. The key is to work with equipment that is solid, is proven, and will most likely remain available for the next several years.
- The user should take the following steps before putting the new system out for bid:
  - Step 1: write the purpose of the proposed CCTV system.
  - Step 2: write the purpose of each camera in the system.
  - Step 3: define the areas to be viewed by each camera.
  - Step 4: Choose a camera style.

- Step 5: choose the proper lens for each camera.
- Step 6: determine the best method for transmitting the video signal from the camera to the monitor.
- Step 7: lay out the control area and determine what enhancements are needed based on each visual assessment point's requirements.
- **High-density video (HDV):** A term used with high-resolution cameras, images, or monitors. HDV does not imply megapixel technology but complies with the international standards of **HDV TV at 1080p (1,080 pixels on a 16:9 screen ratio)**.
- **Megapixel:** A term used to refer to a camera that has an image that is made up of millions of pixel points. Megapixel technology allows for extremely wide views with equivalent 4 CIF or better resolution.

**Resolution.** The number of horizontal scan lines or digital pixel arrays that the camera captures. The identification views.

| View Type                    | Description                                                                                                          | Pixel Resolution                |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------|
| General                      | Clothing and colours not distinguishable.<br>No blowup for detail of person- sized targets.                          | 5 pixels/it.<br>(16 pixels/m)   |
| Monitor                      | General human or vehicular traffic flows.<br>No serious detail upon blowup.                                          | 7 pixels/ft<br>(23 pixels/m)    |
| Detect                       | Person-sized targets large enough to be detected but not identified.<br>No significant detail on blowup.             | 1 pixels/ft<br>(35 pixels/m)    |
| Observe                      | Clothing and colours start to become distinctive.<br>No good detail on blowup of person-sized targets.               | 18 pixels/ft<br>(58 pixels/m)   |
| Recognise                    | High degree of accuracy identifying and separating known individuals.<br>Good detail for general blowup of an image. | 35 pixels/ft<br>(118 pixels/m)  |
| Identification               | Establish identity of individuals beyond shadow of doubt.<br>Excellent image blowup.                                 | 46 pixels/ft<br>(150 pixels/m)  |
| License Plate identification | Identification of license plates and similar-sized objects with excellent detail on blowup.                          | 70 pixels/ft.<br>(231 pixels/m) |

#### 6.6.4. Equipment Selection.

##### ▪ Camera:

CCTV cameras come in four main types.

##### **Standard analogy CCD (Charge Coupled Device) cameras**

- These may be black-and-white or colour.
- Works well in all indoor and outdoor applications.
- They are analogy-based and may or may not have digital effects.
- Resolution ranges from 220 horizontal lines (very low) to 580 horizontal lines.
- Light sensitivity varies between .005 lux (very low) to 10 lux (very high).
- Colour cameras are the most restricted by low-light situations.
- Capturing colour images during daylight hours and black-and-white images at night when the light is low.

##### **IP cameras**

- These digital cameras come in black-and-white or colour.
- These cameras are available in three basic styles: standard, megapixel, smart.
- All IP cameras measure their resolution as a multiple of the Common Intermediate Format (CIF)

- Resolution is 352 x 240 (means 352 pixels on a horizontal basis with 240 pixels on a vertical)
- These cameras have several features such as digital video motion detection, facial recognition, privacy blocking, digital pan/tilt, and zoom

#### **Infrared (IR) cameras**

- These cameras require an IR light source to create an image. They are used where visible light is not an option.

#### **Thermal cameras**

- These require no visible or IR light to produce an image.
- Using special filters and lenses, the cameras monitor the temperatures of the objects in their field of view and use colours to represent temperatures. Cold objects are shown in varying shades of blue, while hot objects are shown in varying shades of red.
- Thermal cameras are often used in long-range surveillance, such as monitoring ships in a harbour five miles out.
- Since these cameras require no light to create an image, they are popular with police and border patrols.

#### ▪ **Lenses**

- After choosing a camera, choosing a lens is the second most important decision of the project.
- Lenses come in five main types:
  - wide-angle
  - standard
  - telephoto
  - zoom
  - varifocal

#### **Wide-angle lens**

A wide-angle lens captures a very wide scene and thus is best suited for short ranges— that is, 0 to 15 ft. (0 to 4.5 m).

#### **Standard lens**

- For an average scene at a medium distance, a standard lens is needed.
- This type of lens **reproduces a view equivalent to what the human eye sees**
- A medium distance would be considered about 15 ft. to 50 ft. (4.5 m to 15.25 m).

#### **Telephoto lens**

- If the required view is a **narrow area at long range**, a telephoto lens is needed.
- A telephoto lens can best be compared to a telescope, as it enables the user to look at objects far away as if they are close.
- However, **such magnification narrows the field of view.**
- Long-range is considered anything **over 50 ft.** (15.25 m).

#### **Zoom lens**

- A zoom lens incorporates moving optics that produce the same views provided by wide-angle, standard, and telephoto lenses, all in one device.
- These lenses may be **manual or motorized.**
- With a manual zoom lens, the installer manually adjusts the lens's field of view and focus. Once set, the lens remains fixed.
- A motorized zoom lens has motors installed within the housing of the lens.

#### **Varifocal lens**

Varifocal lenses differ from zoom lenses in two ways:



- Varifocal length
- Fixed Focal Length

#### ▪ Camera format & Lens

- The resolution of digital cameras is measured in terms of the **Common Intermediate Format (CIF)**.
- A 4-megapixel image has the same resolution as 400 ASA film. A 6-megapixel digital image has a resolution of 100 ASA film.
- The resolution of an image is determined first by the camera, second by the transmission method, third by the weakest link in the video system interface, and fourth by the reproduction capability of the image storage system.

#### • Common IP Resolutions

| <u>Ratio</u>    | <u>Label</u> | <u>Horizontal x Vertical Array</u> | <u>Grid</u>      |
|-----------------|--------------|------------------------------------|------------------|
| CIF             | CIF          | 352 x 240 pixels                   | 84,480 pixels    |
| Quarters of CIF | Q CIF        | 176 x 120 pixels                   | 21,120 pixels    |
| 4 Times CIF     | 4 CIF        | 704 x 480 pixels                   | 3,37, 920 pixels |
| 16 Times CIF    | 16 CIF       | 1280 x 1024 pixels                 | 1.31 megapixels  |
| 25 Times CIF    | 25 CIF       | 1700 x 1200 pixels                 | 2.04 megapixels  |

#### ▪ Controlling Software

- The following recorders are used for recording data:
  - Digital video recorders (DVRs) (analogy).
  - Network video recorders (NVR) (digital).
  - Server/cloud applications (digital).

#### **Digital video recorders (DVRs) (analogy)**

- DVRs capture analogy signals and convert them to digital formats.
- These recorders store video data on a hard drive, CD, DVD, or another medium.

#### **Network video recorders (NVR) (digital)**

- NVRs are designed to accept either digital or analogy signals from the field.
- They are much better than DVRs and are similarly designed to be controllable over a network.
- These recorders also store video data on a hard drive, CD, DVD, or another medium.

#### **Server/cloud applications (digital)**

- These systems offer sophisticated search methods as well as multiple visual outputs for delivery to both fixed and mobile applications.

## 7. Communications and Annunciation Systems.

### 7.1. Alarm Communication & Display.

Transports alarm and assessment information to a central point and presents the information to a human operator.

#### **Critical Elements**

The 2 critical elements of an AC&D system are

- The transportation or communication of data
- Presentation or display of that data to a human operator.

The **Attributes** of AC & D are:

- **Withstand Capacity:** Systems must be able to withstand the environments in which they are placed, and equipment must be designed to withstand without failing.
- **Reliable:** AC&D components must be reliable and have a long mean time between failure (MTBF) as a reliable system requires less maintenance and trust.

- **Backup capability:** By maximizing the robustness, reliability, and redundancy of AC&D systems and taking the chance of failure into account and provide a backup capability.
- **Easy to use:** AC&D systems must be easy for an operator to use.

When an alarm event occurs, the AC&D system must know:

- **Alarm Event:** The operator must know
  1. Where an alarm has occurred
  2. What or who caused the alarm
  3. When the alarm happened
  4. Response or action required.
- **Operator Action:** The operator must know how to respond. This can be accomplished through training and AC&D system prompts.

## 8. Security Communications.

### 8.1. Wire and Cable Communications

8.1.1. Alarm signals may be transmitted on either a shielded or unshielded pair of direct current (DC) conductors. The wire resistance limits the effective length of a line.

#### 8.1.2. Optical Fibre.

- It is a strand of high purity spun glass, typically about the thickness of a human hair. A light source introduces a light beam to a unit at the other end.
- **Use:** Optical fibres carry voice-grade signals, video signals, and digital or data grade signals.
- **Physical Form:**
  - They are much smaller and lighter than conventional wires.
  - They are flexible and can take an irregular course from point to point.

Video Transmission

#### 8.1.3. How Video Transmission done:

- Transmission -> Telecommunication -> Optical Fibber -> Physical Device
- **Transmission:** Video signals transmitted via normal telephone circuits are first converted to digital, then to audio signals.

#### 8.1.4. Status and Alarm Transmission

Three types of line transmission installations used in electronic protection systems:

##### Loop

- **Installation:** Devices installed and looped throughout an area, and connected to the control centre.
- **Usage:** This system is adequate for a **small space or a single facility**

##### Point-to-Point

- **Connection:** Each sensor is connected directly to a control centre by a pair of wires.
- **Price:** This hard-wired installation is more expensive than a loop system but effective.

##### Multiplexing

- It is a technique to transmit several messages simultaneously.
- A multiplex installation can be more cost-effective

The two methods generally used are:

1. Time Division Multiplexing (TDM)
2. Frequency Division Multiplexing (FDM)

- **Time Division Multiplex:** Each sensor or data source is assigned a time segment, and each may transmit only during its assigned segment.

- **Frequency Division Multiplexing:** It signals the number of sensors on a common transmission line over a frequency spectrum.

**Example of FDM:** It is a system where three transducer outputs modulate three subcarrier frequencies

#### 8.1.5. **Wireless Communications**

**Requirement:**

Wireless communication requires:

- Transmitter to furnish radiofrequency energy.
- Antenna to radiate the energy into the atmosphere.
- Receiver.
- Power for the transmitter and receiver.
- **Process** The transmitter modulates carrier wave must create different values, which represent the signal characteristics of the source.
- **Resonate** The receiver should resonate or tunes the signals, amplifies them, and demodulates or detects the original source signals.
- **Modulation** is either **Amplitude Modulation (AM)**, in which the variations are in the amplitude or range of the carrier signal, or frequency modulation (FM).

#### 8.1.6. **Voice Radio**

- **Use:** Voice radio is used in many protection systems.
- **Process:**
- Transmitter Station usually is **located** at the control centre all other alarm system signals terminate.
- Security personnel who require communication with the control centre use mobile or portable units.
- Transmission distances, physical barriers, and signal interference in the area determine the power level and cost of the equipment.

#### 8.1.7. **Wireless Alarm Signals**

- **Consist:** A wireless alarm system consists of an alarm detection array and an RF interface module.
- **Content:** The system vendor provides a network of RF receiver sites, or base stations, which are linked by redundant leased lines to the central station.
- When an alarm sensor is activated, a coded alarm signal is transmitted by an omnidirectional antenna to all base stations within range of the transmitter. The base station receiving the strongest signal transmits the alarm via a leased line to the central station.

#### 8.1.8. **Cordless Telephones**

**Consist:** Cordless Telephone set consists of base and handset transceivers, between which the communication is transmitted by radiofrequency.

**Operating Principle:** The basic operating principle is to provide the mobile user with a radio link.

**Security:** The security of communication can be enhanced using scramblers and is applicable to a cellular conversation.

#### 8.1.9. **Cellular Telephones**

**Digital Transmission Compression Types –**

- Time Division Multiple Access (TDMA)
- Code Division Multiple Access (CDMA)

**TDMA** divides calls into pieces of data that are identified on the receiving end by the time slots to which they are assigned.

**CDMA** spreads segments of calls across a wide swath of communications frequencies these carry a code and they know the originating station.

**Usage:**

Used in:

- In disaster
- In a hostage situation
- Backup to routine security communications

**8.1.10. Private Fixed Wireless Systems****Private Automatic Board Exchange (PABX):****Uses:**

- The low-power transmitter used within a limited range.
- Use in transmission compression technology, such as TDMA

**Satellite Communications**

- **Geo-stationary** earth orbit currently used at high altitudes have an inherent signal delay, which can be a problem in certain communications applications.
- Delay-sensitive communications will benefit from emerging technology that is used in **Low Earth Orbit** (LEO) satellite systems.
- **Medium Earth Orbit** (MEO) satellite systems use satellites 6,500 miles (10,300 km) above the earth.

**9. Scramblers.**

9.1. A scrambler is a tool for disguising information to safeguard it from eavesdropping.

9.2. The type of scrambler to select depends on the importance of the information to be transmitted and the skill of those who might intercept it.

9.3. Voice has two characteristics that may be modified to reduce or destroy its intelligibility— frequency (the pitch of the voice) and amplitude (its loudness). Scramblers invariably distort the frequency content of the voice.

**9.4. Frequency inverters.**

9.4.1. The simplest scramblers are known as frequency inverters. They operate by inverting the frequency content of the voice.

9.4.2. Their major advantage is low cost and tolerance for poor communication channel conditions.

9.4.3. They are inherently a single-code device and can therefore be broken by any listener with a similar scrambler or equivalent.

**9.5. Band splitters.**

9.5.1. Band splitters are extensions of the frequency inverter, in which the single speech band is broken into a number of smaller frequency bands.

9.5.2. These bands are inverted and interchanged prior to transmission.

**9.6. Rolling Band splitters.**

9.6.1. Band splitters may be improved by continuously modifying the way frequency bands are interchanged, in accordance with predetermined pattern.

9.6.2. Can be broken on trial and error basis.

**9.7. Frequency or phase modulators.**

9.7.1. Frequency modulator scramblers cause the voice spectrum to be inverted and continuously changed in frequency in accordance with a predetermined pattern.

9.7.2. Phase modulators operate in similar fashion but change the phase of the voice.

9.7.3. A trained listener may still obtain some intelligence from the scrambled transmission.

**9.8. Masking.**

9.8.1. Masking is a technique for modifying the amplitude of the voice by adding another signal into the voice band.

9.8.2. The masking signal is removed at the receiving unit to restore original voice.

9.8.3. Used in conjunction with a band splitter or frequency or phase modulator, masking provides a high degree of security.

## **10. Alarm Control & Display.**

10.1. The AC&D system's control and display subsystem provides information to a security operator and enables him or her to enter commands affecting the operation of the AC&D system.

10.2. An effective control and display subsystem presents information to an operator quickly and clearly.

10.3. The display subsystem must not overwhelm operators with detail but should show only necessary information.

### **10.4. Ergonomics: Human Factors**

10.4.1. The control and display subsystem should be designed to serve the human operator.

10.4.2. All displays should be roughly perpendicular to the operator's line of sight.

10.4.2. Indications and operator inputs need to be prioritized; the most important ones should be placed in the **primary interface area** and should be within a 30-degree viewing cone.

10.4.3. **Operational displays** that are used often should be placed in the secondary area, where the operator may have to move his or her eyes but not head.

10.4.4. **Support displays** used infrequently, such as backup system and power indicators, may be placed beyond the secondary area.

10.4.5. Different pitches and volumes can be used to distinguish classes of alarms, such as security, safety, or maintenance.

### **10.5. Ergonomics: Graphical Displays**

10.5.1. Well-designed graphical user interfaces (GUIs) can improve the display of security alarm information, and poorly designed GUIs can quickly overwhelm an operator.

10.5.2. An on-screen display window may present text, graphics, or controls. It can be any size, and **up to three windows can be displayed at once.**

10.5.3. One should be the full size of the screen and should contain an overview of the system status.

10.5.4. A smaller window should present subordinate information as needed.

**Subordinate windows should be no larger than half the screen.**

10.5.5. A third window may display menus or other operational controls.

#### **10.5.6. Menu.**

- Menus are typically displayed along the top of a window and can be nested.
- Menus organize system commands clearly and concisely, and their structure should not be overcomplicated.
- A good menu contains no more than **nine items and three levels.**
- Common commands should not be placed in menus but should be available as buttons. Visible buttons should be limited to **nine.**

#### **10.5.7. Maps. scale of maps 1:5,000.**

#### **10.5.8. Colours.**

- Colour can highlight important information but should be used sparingly.
- **No more than seven colours should be used.**



- Maps should be black-and-white or use low-saturation colours. Red, yellow, and green should be reserved to indicate sensor status.

## **Chapter – 12 : Security Officers & Human Element**

### **1. Security Officer Utilisation**

1.1. Security officers are best used as part of the complete protection plan and not as a stand alone resource.

1.2. Security management plans are based on understanding of the risks they are designed to control.

#### **1.3. Number of Officers Required**

1.3.1. The number of officers required for a facility is determined by several factors, including the following:

- Level of risk;
- Physical complexity and size of the facility;
- Number of employees;
- Nature of work;

1.3.2. Assuming that security officers work 40 hours a week, each post that is staffed 24 hours a day, seven days a week, requires a minimum of 4.2 officers. However, to compensate for sick leave, vacations, and holiday leave, each post requires 4.5 officers.

1.3.3. The number of support personnel required depends on the size and complexity of the security force.

1.3.4. The need for security personnel can be analyzed using the "**necessary human being**" concept.

1.3.5. Skills needed: **cognitive (knowledge), psychomotor (physical), and affective (attitudinal) skills.**

1.3.6. A human being is needed if the post requires the ability to do the following:

- Discriminate among events, persons, circumstances, or physical objects based on complex and varying criteria
- Conduct rational dialogue with people face-to-face and discern next actions based on such communication
- Use physical force or the threat of physical force to restrain one or more persons
- Exercise judgment to determine a logical course of mental or physical activity in response to circumstances that may vary randomly
- Provide reports detailing their activities on an ongoing basis

1.3.7. Ultimately, determining the size of the security officer force should be part of a security risk analysis that considers the organization's mission, core values, operating environment, security objectives, threats, vulnerabilities, risk acceptance and other measures.

### **2. Basic Functions.**

#### **2.1. Access Control.**

2.1.1. A **primary function** of the security officer is access control, which is the regulation of movement into, out of, and within a designated building or area.

2.1.2. In times of crisis or unrest, security officers performing access control duties may also be assigned to check individuals' body temperature, watch for inappropriate dress (such as a heavy coat in hot weather), or identify persons behaving erratically.

**2.2. Patrol.**

- 2.2.1. Officers on patrol **observe** a wide variety of people, assets, and locations.
- 2.2.2. Patrolling officers may also be called upon to respond to door alarms and other irregularities.
- 2.2.3. Patrols are generally divided into two categories- foot patrols and vehicular patrols.
- 2.2.4. **Observation** is a prime task for the officer on patrol.

**2.3. Escort.**

- 2.3.1. In general, security personnel should **NOT** be required to escort visitors and customers on company property.
- 2.3.2. In some settings, security officers who have been trained to do so may escort people carrying large sums of money or special information or property.

**2.4. Inspection.** Inspection may be performed along with other assignments, such as patrol. Officers may also conduct special inspections for fire hazards; unlocked doors, safes, and windows and other security concerns.

**2.5. Monitoring.** Security monitoring is conducted from centralized security operations centres or global operations centres. The centralized control centres may be responsible for a single location or may monitor sites across an entire country or countries. Control centre security personnel must be comfortable with electronic systems, computers, and a sometimes high-stress environment.

**2.6. Emergency Response.** Security personnel have a responsibility to be prepared for emergencies. It falls to security officers to direct, coordinate, and carry out the responses established by security management.

**2.7. Dealing with Disruptive People.**

- 2.7.1. Security officers may be called upon to deal with people who appear mentally ill, are under the influence of alcohol or illegal substances, are temporarily unable to control themselves due to physical or emotional trauma, or otherwise behave in an irrational manner that could result in injury to themselves or others.
- 2.7.2. Addressing disruptive behaviour requires sensitivity.
- 2.7.3. The goal is to prevent harm from the person's acts and to control the person with a minimum of injury, discomfort, or embarrassment.
- 2.7.4. The most important consideration is safety and deescalating or eliminating any immediate danger.

**2.8. Observe & Report.** By being observant and understanding what they are looking for, security officers play a key role in the prevention of unwanted incidents,

**2.9. Public Relations.**

- 2.9.1. Security officers play a public relations role when they perform their protection duties and represent an employer.
- 2.9.2. They are often the first and last contact a visitor, customer, vendor, or employee has with an organization.

2.9.3. Security personnel must ensure that rules are followed, but they must also treat employees and visitors in a fair and understanding manner. **Uncooperative** employee should be referred to a supervisor for administrative action.

### 3. Weapons.

#### 3.1. **Firearms.**

3.1.1. The decision whether to arm an officer should be based on the existence of one or both of the following conditions:

- There is a greater danger to life safety without the weapon.
- The officer may reasonably be expected to use fatal force.

3.1.2. If officers are armed, the management of the facility assumes several responsibilities:

- Proper training of the officers to be armed;
- Selection of the appropriate firearms and ammunition;
- Proper maintenance of the firearms by a qualified gunsmith;
- Maintenance of records of the foregoing actions; and
- An adequate level of liability insurance.

3.1.3. Firearms may be appropriate when large amounts of cash are regularly handled and robbery is a risk.

3.1.4. The strategic value of issuing weapons is that potential assailants may be dissuaded from attacking a target if they know they will face armed resistance.

#### 3.2. **Less than Lethal Weapons**

3.2.1. A full analysis of "will the benefits outweigh the risks" should be conducted.

3.2.2. Listed below are the most common types of less than lethal weapons.

- **Electroshock weapons.** These defensive weapons use electricity to stun or disable a subject. There are three common types: the TASER, the stun gun, and the stun baton.
- The **TASER** (a registered trademark) uses compressed gas to fire two barbed probes towards a target. The subject's neurological system is affected, and the target will generally lose control of muscles.
- Using a **Stun Gun** requires the officer to directly contact the subject with the device. Stun guns and stun batons may cause pain without disabling the subject. Also known as conducted electrical weapons (CEW) or conducted energy devices (CED), these weapons' effects are temporary, but may stop aggressive action.
- **Batons.** Perhaps one of the most utilized less than lethal weapons in security is the baton. Batons may be composed of wood, metal, or a plastic like resin. Improper use of a baton can cause great bodily injury and/or death.
- **Aerosol irritants.** The carrying of aerosol irritants or chemical agents by security officers is common. CS (2-chloroacetophenone), CN (0-chlorobenzylidene), and OC (oleoresins capsicum) are among the most frequent chemical compounds used. Commonly referred to as "**tear gas**" or "**pepper spray**," they are designed to irritate the mucus membranes and the eyes causing tearing, coughing, difficulty breathing, and skin irritation.

#### 3.3. **Policy Considerations.**

3.3.1. **Use-of-Force Policy.** A comprehensive use-of-force policy: (UNODC 2020)

- Identifies the circumstances and limitations that govern use of force, including:
  - Use of force in individual self-defence

- Use of force to protect others
- Use of force to protect property, and response to serious crime
- Identifies the limits of the use of force acceptable to the government and organization.
- Can be used to determine whether a particular use of force is reasonable and necessary

**3.3.2. Police security force continuum.** A concept that is taught at most police academies and includes six broad levels.

- **Level One: Officer Presence.** A visible uniformed security officer or a marked vehicle is often enough to deter criminal or suspicious activity. At this level, **gestures should be nonthreatening.**
- **Level Two: Verbal Communication.** In combination with a visible presence, the spoken word can increase deterrence. The officer should begin with a calm but firm message with the goal of de-escalating the situation.
- **Level Three: Control Holds and Restraints.** Using bare hands to guide, hold, or restrain an Individual does not include offensive moves such as punching, tackling, or choking.
- **Level Four: Chemical Agent.** These defensive weapons may be used only when lesser tactics have failed or when a physical threat is imminent.
- **Level Five: Temporary Incapacitation.** When the situation is so extreme, violent, and immediate that it is necessary to temporarily incapacitate a suspect, nonlethal force from an empty hand to an intermediate weapon such as pepper spray, TASER, or baton may be used.
- **Level Six: Deadly Force.** Only in cases where there is immediate fear of death or great bodily injury at the hands of a perpetrator is deadly force acceptable.

#### **4. Uniforms & Equipment.**

##### **4.1. Uniform.**

4.1.1. Most organizations supply security officers with uniforms and equipment. Doing so ensures that the uniforms are, in fact, uniform in appearance and properly fitted to each officer.

4.1.2. Company policy should address whether uniforms should be worn off duty, including travel to and from work.

##### **4.2. Body worn Camera.**

4.2.1. Organizations that use the cameras cite their value in increasing accountability and providing evidence. Those who do not use them cite the cost of video storage, and the consequences when the cameras do not collect video due to malfunction or misuse.

4.2.2. To address privacy concerns, officers using the cameras must let members of the public know they are being recorded.

#### **5. Training Program Assessment & Design.**

##### **5.1. Behavioural Objectives.**

5.1.1. Based on the identified training need, behavioural objectives should be written for the actual training program.

5.1.2. Each objective should describe the desired task, the conditions for performing the task, and the criteria for an acceptable performance of the task.

5.1.3. Since the objectives guide the development of instruction and tests, they can be most effective when written with the help of a consultant.

5.2. **Designing the Instruction.**

5.2.1. The key aspects are the ability to design the instruction and then to prepare tests and documentation.

5.2.2. Developing training : Internal team, external team or purchasing off-the-shelf

5.2.3. **Certified Trainer Syndrome**, which can occur when an instructor purports to be certified on a topic but has no background to teach a class.

5.3. **Training Evaluation.**

5.3.1. The number or proportion of employees exposed to training is considered the learning contact.

5.3.2. When evaluating training, the learning contact is important because programs that the entire security staff does not participate in are of limited, and perhaps questionable, value.

5.3.3. Participant evaluations are usually not good as validation tools. however, because they do not measure actual learning. Rather, they validate what the learners thought they learned.

5.3.4. Tests should be as objective as possible: the respondent either supplied the correct answer or did not.

5.3.5. Test design is one part of training development where external experts should be consulted.

6. **Training Methodology.**

6.1. **Lectures.**

6.1.1. A traditional method used to instruct personnel. Lectures are appropriate for introducing a topic and setting the course of the instruction.

6.1.2. Lectures are not appropriate for individualized instruction.

6.2. **Case Studies.**

6.2.1. The case study method is useful for guiding the learner in any topic where discretionary judgment is necessary.

6.2.2. Case studies can be used to supplement lectures and give them life.

6.2.3. Case study method works best when students work together in groups where there is active discussion.

6.3. **Job Aids.**

6.3.1. Job aids are convenient reminders on how to do a task.

6.3.2. Also called "takeaways, job aids help ensure that correct procedures presented previously to officers in training sessions are being followed.

6.3.3. They are a supplement-not a substitute-for training.

6.4. **Distance Learning.**

6.4.1. Instruction where the learner and teacher are at different locations but are still able to communicate is called distance learning.

6.4.2. can be delivered in a number of formats, from the old-fashioned correspondence course to computer-based instruction.

6.4.3. College research papers could be considered a form of distance learning.



**6.5. On the Job Training.**

6.5.1. On-the-job training (OJT) is easy to develop. It is an especially useful way to ensure that transfer of learning, from instructor to student, has occurred.

6.5.2. OJT works well when it is structured with specified procedures.

**6.6. Mentoring.**

6.6.1. In a mentoring program, a new security officer, after completing pre-assignment training, is placed with a mentor who coaches him or her on the job.

6.6.2. Bridge any gaps between classroom-based pre-assignment training and the realities of the job environment.

6.6.3. The mentor teaches but does not evaluate.

**7. General Orders.**

7.1. General orders may be thought of as canons or bodies of principles for protection officers.

7.2. Some commonly used general orders are as follows:

- Familiarity, fraternization, and unnecessary conversation with persons while on duty are prohibited.
- Officers will maintain a military or professional bearing.
- Smoking and eating on duty are prohibited. •
- All statements to the media will be made by the designated media spokesperson or public information officer (PIO).
- Security officers must enforce regulations diplomatically.
- Officers shall wear the employer's uniform and use the employer's equipment only when and where authorized.
- Officers will promptly report any criminal conviction to their employer.
- Officers will remain at their post unless properly relieved or otherwise directed.
- Officers leaving a post will brief their relief fully and pass along any orders that have been issued.

**8. Post Orders.**

8.1. Post orders are the most important written instructions for the security force. Post orders do the following:

- Express the policies of the protected enterprise.
- Summarize required officer duties.
- Avoid the problems of word-of-mouth instructions.
- Provide a basis for site-specific training.

8.2. Clear and understandable post orders are important in serious incidents that may call into question the integrity, competence, or capacity of an officer.

8.3. Post orders should be developed with the following criteria in mind:

- Each order deals with a single subject.
- Each order is as brief as possible.
- Each order is written in simple, easy-to-understand terms.
- Orders are indexed in detail.

8.4. In a fixed post, post orders should be stored at a designated location.

8.5. For a patrol post in a vehicle, the post orders should be carried in the vehicle.

8.6. Each set of post orders should be reviewed periodically to ensure that the orders are current and complete.

**9. Supervision.****9.1. Supervision.**

9.1.1. The quality of assets protection is a direct function of local training and local supervision.

9.1.2. Training occurs when a supervisor observes the ongoing performance of an officer and provides feedback in response to situations that were not correct or where the officer appeared unsure or confused.

9.1.3. Other training is provided when new material is introduced that requires familiarization by the individual officer or the entire shift.

9.1.4. Training also occurs when new security personnel are introduced to the site.

**9.2. Evaluation.**

9.2.1. The assessment must include at least the following items.

- Personal appearance and condition of the officer;
- Physical condition of the post;
- Availability and condition of all required personnel and post equipment, including the post orders;
- Quality of officer response to training questions or situations; and
- Quality of officer response to actual situations arising at post during the visit.

**Chapter -13 : Principles of Project Management****1. Project Management.**

1.1. Project management involves planning, organizing, and controlling resources on a project.

1.2. A project differs from ordinary operations in that projects are temporary efforts with a defined beginning and end.

1.3. Projects are undertaken to meet specific goals and objectives, typically to benefit the organization in some very specific way.

1.4. Projects typically are limited in both time and budget. They have a start and finish date and an allocated budget.

1.5. All security projects work within a set of constraints:

1.5.1. project scope

1.5.2. project schedule

1.5.3. project budget

1.5.4. project quality

1.6. Projects are administered across a series of phases, which typically include the following:

1.6.1. project conception

1.6.2. Project planning

1.6.3. Project design management

1.6.4. Project bid process management (or participation)

1.6.5. Project construction (or construction review)

1.7. The project manager's tools include these:

1.7.1. training (project management is not a natural skill; it must be learned)

1.7.2. project management software

- scheduling software

- office software

1.7.3. determination (forcefully fighting the necessary battles during the project to control scope, schedule, and budget)

## **Chapter -14 : Project Management – Systems Design**

### **1. Systems Design Principle.**

- 1.1. System design starts with planning and assessment.
- 1.2. This comprises of collecting information on security needs, objectives, constraints so that risk management and control can be affected before the security event.
- 1.3. The second task of planning & assessment phase is to analyse security requirements and formulate solutions or countermeasures.
- 1.4. The output of the design phase is systems hardware, software identification and placement, integration, and performance documentation that is sufficiently clear and complete to ensure consistent, accurate interpretation by suppliers and installers for systems procurement and implementation.
- 1.5. System design is a **serial process**. Each phase and task must be performed before the next can begin; the output from one task becomes the input for the next.

### **2. Planning & Assessment Phase.**

- 2.1. This phase consists of gathering all relevant pre-design asset information and analyzing it in terms of project requirements and constraints.
- 2.2. Planning and assessment efforts always culminate in a security basis of design, the first and most important output of the design process.
- 2.3. Three key ingredients in the planning phase determine its eventual success.
  - 2.3.1. First, a multi-disciplinary and committed approach from either a single individual or project team is needed.
  - 2.3.2. Second, spending the necessary time and effort in the planning phase results in a more accurate and responsive design solution, reduced risks, reduced overall costs of potential losses, and increased longevity and effectiveness of the installed systems.
  - 2.3.3. Third, decisions made during the planning and assessment phase must be made on the basis of sound and relevant risk and asset environmental information.
- 2.4. The outcome of the overall planning phase is a set of security requirements or objectives that is used as a basis of the eventual design (**also called design basis**).
- 2.5. Another important outcome of the planning phase is the development of the business case for the new or upgraded security systems.
- 2.6. The business case documents the impact of the design solution on the business, the necessary investments, expected quantifiable savings, and other metrics that allow decision makers to make investment decisions on a security project.
- 2.7. A requirements analysis is necessary for effective planning. **Requirements analysis uses the threat, assets, and risk analysis as its basis.**
- 2.8. **The level of protection for a group of assets must meet the protection needs of the most critical asset in the group.**

### **3. Design Phase.**

#### **3.1. Development of Design Criteria.**

- 3.1.1. Constitutes the ground rules and guidelines of design.
- 3.1.2. **Codes & Standards.** Applicable codes must be identified and applied to the initial design to ensure compliance.

3.1.3. **Quality.** A good design always strikes a balance between quality components and overall cost.

3.1.4. **Capacity.** Planning for 10- 15 % spare capacity.

3.1.5. **Cost.** Two of the main cost drivers for systems implementation are the design fees and projected system construction costs.

3.1.6. **Operations.** Two main criteria drive security designs, First, security programs need to have minimum negative impact on productivity and facility operations. Second, security operations should be seen as a natural use of security systems.

3.2. **Basis of Design.** The basis of design becomes the designer's means to obtain consensus from the design team on the goals and objectives of the project.

3.3. **Conceptual Design.** The conceptual design also called a design concept or schematic phase, is the last task of the planning and assessment phase and is initially developed as a product of risk assessment.

3.4. **Design & Documentation Phase.**

3.4.1. In the construction design industry, this may be split into two phases, the design development phase and construction documents phase.

3.4.2. The objective of the design and documentation phase is to complete the design and to document the process to the level of detail necessary for the chosen method of procurement.

3.4.3. This phase includes deliverables such as specifications, evaluation criteria, and implementation schedule.

3.4.4. The complete set of procurement documents, known as contract (or bid) documents, will consist of three sections: **contractual details, construction specifications, and construction drawings.**

3.4.5. In a construction-related procurement the specifications and drawings are called construction documents.

3.4.6. **Design Development.**

- Design development (DD) drawings typically include a title sheet, list of proposed schedules (spreadsheets) for each system (power, heat load, wire types, etc.), individual floor plans showing proposed device types and placements, and a list of physical details and single-line (block) diagrams to be included in the construction documents package.
- The title sheet typically includes a drawing list, symbols list, abbreviations list, general notes, and other schedules that are common to all drawings in the package.

3.4.7. **Construction Documents.**

- A typical CD phase package may include the following:
  - CD drawings
  - Specifications
  - Bill of quantities
  - Refined system budgets
  - Contract terms and conditions (usually from the owner's purchasing department)
- CD drawings differ from DD drawings in their level of detail.

3.8. **Specifications.** The systems specifications mirror and complement the actual systems design in sufficient detail to achieve the following:

- The final implementation reflects what was intended.
- All bidders get the same, complete understanding of the requirements.

3.9. **Drawings.**

3.9.1. Drawings are the cornerstone of any system implementation.

3.9.2. Security system drawings usually consist of plans, elevations, details, risers, and hardware schedules.

### 3.9.3. Plans.

Each system's implementation plan shows a top-down, map-like view of an area where security devices and systems are located.

The background information on a plan drawing consists of such items as fence lines, building wall locations, interior partitions, doors, furniture, door and room numbers (known as targets or tags), room names, floor materials, stairs, fixed equipment, etc.

The most common symbols set for manual drafting is issued by ASTM.

**3.9.4. Elevations.** Elevations are views of vertical surfaces and are included to show mounting heights and locations of wall-mounted devices, such as cameras, card readers, and motion sensors.

**3.9.5. Details.** Detailed drawing sheets can be developed to define elements of the system in more detail.

**3.9.6. Risers.** Riser diagrams are representations of complete subsystems and they schematically demonstrate all the associated devices and components and their interconnecting cables.

### 3.10. Hardware schedules.

**3.10.1.** Hardware schedules for system implementation are tables of related security devices.

**3.10.2.** They provide detailed information that cannot easily be shown on drawings or in the text of a specification.

**3.10.3.** Schedules are used for door hardware, control devices, intrusion sensors, cameras, monitors, and other devices that appear repetitively, such as termination panels.

**3.10.4.** *The schedules are often shown on security drawings but may also be appended to security system specifications.*

## 4. Contracting.

**4.1. Initial Budget.** An additional product of the planning phase is the initial budget, both for capital expenditures and recurring costs associated with the proposed system.

### 4.2. Types of Cost Estimates.

#### 4.2.1. Budgetary Estimates.

- Budgetary estimates are prepared during the initial planning phase for a new security system.

- The goal is to arrive at a cost figure that can be used for getting the new security system into the budget cycle.

- These estimates are used for budgetary purposes; they have a large contingency, such as plus or minus **10 to 20 percent**.

#### 4.2.2. Preliminary Design Estimates.

- If the security system project is part of a larger construction project, the process may require a preliminary design estimate.

- This estimate should be developed at the 50 percent design review stage and normally has a contingency of plus or minus 10 percent.

#### 4.2.3. Final Design Estimates.

- The estimate is refined as the project advances to 100 percent completion.

- This estimate should have minimal contingency, on the order of plus or minus 5 percent.

**4.3. Life Cycle Cost.** The actual cost of a security system is its life-cycle cost. Life-cycle cost estimates include the following components:



4.3.1. **Engineering and design costs.** These are the costs associated with the planning and design of the security system.

4.3.2. **Hardware.** The hardware costs include the original equipment plus start up spare parts.

4.3.3. **Software.** The software costs are for the operating system and application system software necessary to operate the security system.

4.3.4. **Installation costs.** Installation costs include labour expended in installing equipment and software, labour to perform inspection, testing and commissioning, equipment rental, bonding, supervision, and overhead.

4.3.5. **Operating costs.** Operating costs include expenses for personnel, power consumption, and consumables (such as paper and ink cartridges).

4.3.6. **Maintenance costs.** Maintenance costs include labour and spare parts for preventive and remedial maintenance.

4.3.7. **Other costs.** Other costs include state and local taxes, profit (10 percent), performance bonding (3-5 percent), and contingency (5-10 percent).

4.3.8. **Adjustments.** RS Means data is based on national averages. For specific locations, the cost data may need to be adjusted.

## 5. **Procurement.**

### 5.1. **Procurement Forms.**

5.1.1. The three major forms of security systems procurement are sole source, request for proposal (RFP), and invitation for bid (IFB), with some variations depending on whether the buyer is a government agency or commercial firm.

5.1.2. The form of procurement should be selected before or at the start of the design phase.

#### 5.1.3. **Sole Source Procurement**

- For small projects, this may be the most appropriate method of procurement.
- The organisation prequalifies a reputable contractor.

#### 5.1.4. **Request for Proposal**

- The RFP is the most common method of procurement and almost always based on a set of detailed design and construction documents on the basis of the organization's functional requirements.
- The specifications are usually generic and performance-based.
- Equipment makes and models are often listed with the phrase "or approved equal."
- An RFP response may be open to any contractor or it may be limited to a list of prequalified contractors.
- It is common to allow responders to propose alternative solutions, called alternates.
- The RFP need not restrict the organization to accept the lowest bid. Instead, it aims to obtain the best value.
- Three to four weeks is the typical minimum proposal preparation time for medium to large projects.

#### 5.1.5. **Invitation to Bid.**

- IFB is commonly used by government and other organizations whose procurement procedures require that projects be competitively bid and that the award be given to the lowest qualified, responsive bidder.
- No technical proposals or alternative solutions are sought, so the construction documents must be extremely explicit to include the cost of materials and installation.
- The onus of selecting equipment makes and models, and the accuracy of the security system design, is placed solely on the design team.

- Bidders submit a cost proposal or bid, which may contain unit pricing and whatever price breakdown is requested.
- The award is then made, usually without negotiation, to the lowest qualified bidder who has conformed to the bidding instructions.

## **5.2. Procurement Process.**

### **5.2.1. Pre Bid Conference.**

- A pre-bid conference is held in which a representative of each contractor is invited.
- At that conference, the owner or the owner's consulting engineer provides a complete review of the bid documents and a walk-through of affected buildings and locations.
- If applicable, the conference can be held at the site where the new security system will be installed so that bidders can see the field conditions.
- The conference should be held approximately one week after the construction documents have been issued for bid.
- All questions and answers at the conference should be recorded by a design team representative in the meeting minutes.

### **5.2.2. Verification & Evaluation of Bids.**

Once contractor proposals or bids have been received, they are checked for **completeness and accuracy**.

When comparing proposal costs, the life-cycle cost of each proposed system should be calculated.

In its simplest form, ***the life- cycle cost is the sum of the capital cost and the maintenance cost over the useful life of the system.***

The maintenance and warranty costs equal **11 percent** of the total capital systems construction cost.

**5.2.3. Price Negotiation.** Negotiating the final price with the short list of contractors, if permitted by the procurement regulations, should be done on the basis of value.

## **Chapter -15 : System Implementation**

### **1. System Implementation.**

1.1. Project and system implementation begin immediately following the selection of a contractor.

1.2. Steps involved : ***site preparation, contractor coordination, installation, testing and warranty issues, training.***

### **2. Site Preparation.**

2.1. Most important step in installing physical protection system elements is site preparation and planning.

2.2. The contractor should verify the locations and note any changes needed.

2.3. The contractor should visit the site and verify that conditions agree with the design package, prepare a written report and not to take any corrective action without written permission from the customer.

2.4. The contractor inspect, test, and document all existing physical protection equipment and signal lines that will be incorporated into the new system.

3. **Contractor Coordination.** The contractor should constantly coordinate the resources (people and material) with the environment and the project schedule.

#### 4. Installation.

##### 4.1. **Installation & Operation.**

4.1.1. The project manager should instruct the contractor on installing all system components, including any customer-furnished equipment.

4.1.2. After installation, the project manager can tune the system to the specific operations of the facility.

4.2. **Component Installation.** Details on installing physical security components legislated in a standard from the **National Fire Protection Association, NFPA 731**, Standard for the Installation of Electronic Premises Security Systems (2020).

##### 4.2.1. **Card Readers.**

- Card readers should be suitable for surface, semi-flush, pedestal, or weatherproof mounting as required.

##### 4.2.2. **Electric Door Strikes or Bolts.**

- **Solenoids.** Should use direct current (DC) to energize the solenoids. The actuating solenoid for the strikes or bolts should not dissipate more than 12 watts and should operate on 12 or 24 volts DC. Should incorporate end-of-line resistors to facilitate line supervision by the system.

##### 4.2.3. **Electromagnetic Locks.**

- Electromagnetic locks should contain **no moving parts** and depend solely on electromagnetism to secure a portal, generating at least **1,200 lb. (544 kg) of holding force**.

- Release automatically in case of power failure.

- Incorporate an end-of-line resistor to facilitate line supervision by the system.

##### 4.2.4. **Bell or Alarm Box**

- Should be mounted on the front of the facility or in a location where it will be in full view of neighbours and passers by. The alarm should be placed high enough on the building to be out of easy reach.

##### 4.2.5. **Control Panels.**

- Located internally, in a secure location.

- They should be positioned in a manner that it cannot be reached without a ladder.

- Should be close to a main electricity supply.

- Should be locked and have 24/7 alarm supervision using tamper switches.

##### 4.2.6. **Passive Infrared (PIR) and Dual-Technology Motion Detectors**

- Standard PIR motion detectors should **not be mounted** where they might be exposed to infrared light to **avoid nuisance alarms**.

- **Microwave detectors can sense motion through thin walls.**

##### 4.2.7. **Shock Sensors**

- Usually fitted to areas susceptible to forced entry, such as door or window frames.

- Door contacts detect the opening of a door or window but not breakage.

- Shock sensors or PIR detectors used to detect attempts to gain access by kicking a panel out of a door or breaking a window.

##### 4.2.8. **Interconnection of Console Video Equipment**

- For digital video systems using a TCP/IP backbone, the run length of any single connection **should not exceed 328 ft. (100 m)**.

- The signal path between a patch panel and PoE network switch should be **less than 10 ft. (3.048 m)**.

- Between analog video equipment, the contractor should connect signal paths of 25 ft (7.6m) or less with RG-59/U coaxial cable.

#### 4.2.9. Surveillance Cameras.

- Aim the camera to cover the protected area of interest.
- For a fixed-mount camera installed outdoors and facing the rising or setting sun, aim the camera sufficiently below the horizon that the camera will not directly face the sun.
- Focus the lens to give a sharp picture over the entire field of view.
- **Chapter 7 of NFPA 731 provides details on selecting the appropriate location and lenses for cameras.**

#### 4.3. Other Features & Considerations.

##### 4.3.1. **Conduit**

- All interior wiring-including low-voltage wiring outside should be installed in rigid, galvanized steel conduit conforming to applicable standards.
- Minimum conduit size should be **1/2 in. (1.3 cm)**.

4.3.2. **Grounding.** Must be in accordance with NFPA 70, National Electrical Code (2011), articles 250 and 800 (in the United States) or other applicable national standards.

4.3.3. **Cold Galvanizing.** All field welds and brazing on factory galvanized boxes, enclosures, and conduits should be coated with a **cold galvanized paint containing at least 95 percent zinc by weight.**

#### 4.4. Tuning the Alarm System .

4.4.1. After installation, the system must be tuned to the operation of the facility.to avoid generating unwanted nuisance.

4.4.2. Tuning the system requires knowing how the facility operates, what employees come and go, and what types of activities take place.

##### 4.4.3. **Time Periods for Alarms.**

- To tune the system, the security manager should periodically run system reports and look at the alarm history, which shows nuisance alarms and alarm location, frequency, and timing. Patterns may emerge.

#### 5. **Testing & Warranty.**

5.1. The ideal **acceptance tests** stress the system up to the established limits of site-specific threats. Tests should simulate actual threat conditions and provide conclusive evidence about the effectiveness of the security system.

5.2. **Equipment performance** testing is designed to determine whether equipment is functional, has adequate sensitivity, and will meet its design and performance objectives. Equipment performance tests must always be coordinated with appropriate facility personnel and may **require special test equipment.**

5.3. **Personnel performance** tests are intended to determine whether procedures are effective, whether personnel know and follow procedures, and whether personnel and equipment interact effectively.

#### 5.4. Predelivery or Factory Acceptance Test.

5.4.1. The customer may require the contractor to assemble a test system including some or all of the system components, and then conduct tests to demonstrate that system performance complies with specified requirements.

5.4.2. The tests should be scheduled in advance of any installation of the new system, and the customer should attend and observe the tests.

5.4.3. Original copies of all data produced during factory testing, including results of each test procedure, should then be delivered to the customer.

### 5.5. Site Acceptance Testing.

5.5.1. The customer should require the contractor/integrator/installer to develop a plan to calibrate and test all components, verify data transmission system operation, install the system, place the system in service, and test the system.

5.5.2. Before conducting the site testing, the contractor should provide a report to the customer describing results of functional tests. This report should be received at least **two weeks before the start of site testing** and should also include a copy of the approved site acceptance test procedures.

5.5.3. The site acceptance test should be started after written approval has been received from the customer.

5.5.4. The contractor should be instructed that the customer may terminate testing any time the system fails to perform as specified.

### 5.6. Reliability or Availability Testing.

- Reliability testing is best conducted in alternating phases of testing and evaluation to allow for validation of the tests and corrective actions.
- The reliability test should not be started until the customer notifies the contractor, in writing.
- The contractor should provide one representative to be available 24 hours per day, including weekends and holidays (if necessary), during reliability testing.
- The customer should terminate testing whenever the system fails to perform as specified.

- **Phase I Testing.** The reliability test should be conducted **24 hours per day for 15 consecutive calendar days**, including holidays, and the system should operate as specified. If the system experiences no failures during Phase I testing, the contractor may proceed directly to Phase II testing after receipt of written permission from the customer.

- **Phase I Assessment.** The report should explain in detail the nature of each failure, corrective action taken, and the results of tests performed; it should also recommend when to resume testing. **A week after** receiving the report, the customer should convene a **test review meeting** at the job site to discuss the results and recommendations.

- **Phase II Testing.** Phase II testing should be conducted **24 hours per day for 15 consecutive calendar days**, including holidays, and the system should operate as specified.

- **Phase II Assessment.** The report should explain in detail the nature of each failure, corrective action taken, and the results of tests performed; it should also recommend when to resume testing. **A week after** receiving the report, the customer should convene a **test review meeting** at the job site to discuss the results and recommendations.

### 5.7. Post Implementation Testing.

5.7.1. **Operational tests.** Operational tests are performed periodically to prove correct system operation but do not involve verification of equipment operating specifications.

5.7.2. **Performance tests.** Performance tests verify that equipment conforms with equipment or system specifications thereby demonstrating effectiveness.



5.7.3. **Post-maintenance tests.** Post-maintenance tests are operational tests conducted after preventive or remedial maintenance has been performed on security systems to make sure the systems are working properly and according to specifications.

5.7.4. **Subsystem tests.** Subsystem tests ensure that large parts of the system are all working together as originally designed.

5.7.5. **Limited scope tests.** Limited scope tests are used to test a complex system that is broken down into several subsystems or segments that are tested separately.

5.7.6. **Evaluation tests.** Evaluation tests are periodic, independent tests of the security system to validate the vulnerability analysis and ensure that overall effectiveness is being maintained. **An evaluation test should be performed at least once a year.**

## 5.8. Warranty Issues.

5.8.1. The contractor should provide a warranty that is required to repair, correct, or replace any defect for a **period of 12 months**.

5.8.2. The common time for the contractor to report to the job site to address a warranty issue is **within four hours**.

5.8.3. Waiting for manufacturers to replace or repair equipment is not acceptable.

## 6. Training.

6.1. Approval of training content & schedule should be obtained at least **30 days** before training.

6.2. The **first training class** should be scheduled so that it is completed about **30 days before factory acceptance testing** (if conducted) or **site acceptance testing**.

6.3. A **second training** class should be conducted **one week before the start of acceptance testing**, and the system administrators should participate in the acceptance tests and reliability testing.

6.4. **System Monitoring Training.** This training focuses on day-to-day system operation. The training class should be **conducted before the acceptance test**, and the system operators should participate in the acceptance tests and reliability tests.

6.5. **System Troubleshooting and Maintenance Training.** This is usually conducted by a security service provider or integrator.

## Chapter – 16 : Follow- on and Support Activities

### 1. Follow on & Support Activities – General.

1.1. It is essential that security systems be selected based on the support they can receive from the local market after installation.

1.2. The essential elements of a successful security system maintenance program: **maintenance, evaluation, and replacement.**

### 2. Maintenance.

#### 2.1. Remedial Maintenance.

2.1.1. This corrects faults and returns the system to operation in the event that a hardware or software component fails.

**2.1.2. Maintenance Plan.**

- Hardware and software system maintenance may be done by the equipment manufacturer, a system integrator, a maintenance contractor, the users, or any combination thereof.
- Periodic tuning of the security system to each facility to eliminate nuisance and false alarms that create problems for the personnel monitoring and responding to the system.

2.1.3. **Service Level.** Service levels and costs depend on the location of the system in relation to the supplier and on the ability to diagnose and fix problems remotely.

**2.1.4. Prices & Payments.**

- The fees may be arrived at from a mix of factors, including the complexity of the security system, the cost of spare parts, the estimated number of failures per annum, the product usage frequency, the number of security system users, and the age of the security system.
- For high-volume or standard systems, the fee may simply be a set percentage of the purchase price.
- Support fees may also be affected by the geographic location of the system or the ability for online diagnosis and support.
- Not typically covered in a maintenance agreement are such items as misuse, vandalism, lack of training due to turnover, acts of God, etc

**2.2. Preventive Maintenance.**

2.2.1. This consists of scheduled maintenance to keep the hardware and software in good operating condition. the best solution is to select a single contractor/integrator to take responsibility for the maintenance requirements of the system.

2.2.2. Checklists should be developed to ensure that preventive maintenance tasks are performed adequately.

2.2.3. To conserve travel time, preventive maintenance activities should be pursued simultaneously with remedial maintenance activities to the extent possible.

**3. Replacement.**

3.1. To justify the replacement cost, the security manager should consider such factors such as the cost of maintenance, lack of spare parts, obsolescence of hardware and software, operating costs, and unreliability.

3.2. Replacement may also be justified by new technologies and features that provide improved security, the ability to reduce manpower, or other benefits.

## **Physical Security & Life safety in High Rise Buildings**

**1. High Rise Structure.**

1.1. Considerations for High rise structure.

1.1.1. Extends higher than the maximum reach of available fire-fighting equipment.

1.1.2. Between **75 ft. (23 m) and 100 ft. (30 m)**

1.1.3. Approximately **7 to 10 stories.**

1.2. The exact height above which a particular building is deemed a high-rise is specified by the fire and building codes in the area in which the building is located.

1.3. The world's first high-rise structures were built in the United States around the end of the 19<sup>th</sup> century.

## 2. Life Safety Considerations.

2.1. For a high-rise structure the risks and potential losses are higher due to the large size of occupied floor areas, potential difficulties in responding to and containing threats, and the inability to effect immediate evacuation of an entire building.

2.2. The concepts for life safety protection are: planning must address identified assets at risk, and solutions should follow the principles of deter, detect, delay, and respond.

2.3. The most critical threats in high-rise structures include fire, explosion, and contamination of life-support systems such as air and potable water supplies.

2.4. The most significant factors affecting life safety in high-rise structures are these:

2.4.1. Early detection and precise location of incipient hazards

2.4.2. Reliable communications : Internal & external agencies

2.4.3. Assurance of safe escape routes

2.4.4. Prompt application of appropriate control measures

## 3. Considering Building Vulnerabilities.

3.1. Various threats to high-rise structures can spread upward, downward, or horizontally.

3.2. Fire, can spread to higher floors because of the natural tendency of heated gases to rise.

3.3. Contaminants in the air supply, such as smoke or chemical or biological agents, can travel with the air from the point of entry through its distribution pattern.

3.4. Contaminants in the water supply move with the water from the point of introduction.

3.5. Communications facilities within a high-rise are usually routed through common **vertical risers** for the entire height of a structure and distributed horizontally from junction or service ports on each floor.

## 4. Fire Detection.

4.1. The detection of products of combustion or the combustion process-is the first step in any response.

### 4.2. **Smoke Detectors.**

4.2.1. Smoke detectors are automatic fire detection devices designed to detect the presence of smoke.

4.2.2. Smoke detectors generally are **located in open areas**, spaces above suspended ceilings, spaces under raised floors.

### 4.3. **Heat Detectors.**

4.3.1. Heat detectors are automatic fire detection devices designed to sense a certain temperature or rapid change in temperature.

4.3.2. Heat detectors are very reliable and have the **lowest false alarm rate** of all automatic fire detectors.

4.3.3. They are **best suited for fire detection in small confined spaces** where rapidly building high-heat output fires are expected.

### 4.4. **Sprinkler & Stampede Alarms.**

4.4.1. A combination of underground and overhead piping that is connected to an automatic water supply and is installed throughout the building and is activated by heat from fire and discharges water over the fire area

4.4.2. A **standpipe system** in a high-rise building is designed to transport water vertically to upper floors of the building so that a fire can be fought manually with hoses.

5. **Building Operating Modes.**

5.1. "**Open**" buildings. Access is typically **unrestricted** at the building entry level. Tenants, building employees, and visitors proceed directly to their destination floor, via building elevators, where the occupier of that space determines the level of entry control and security.

5.2. "**Closed**" buildings. Access to elevator banks at the building entry levels is **controlled** by some form of credential checking, visitors are authenticated prior to proceeding to elevators, and sometimes a separate messenger centre is provided.

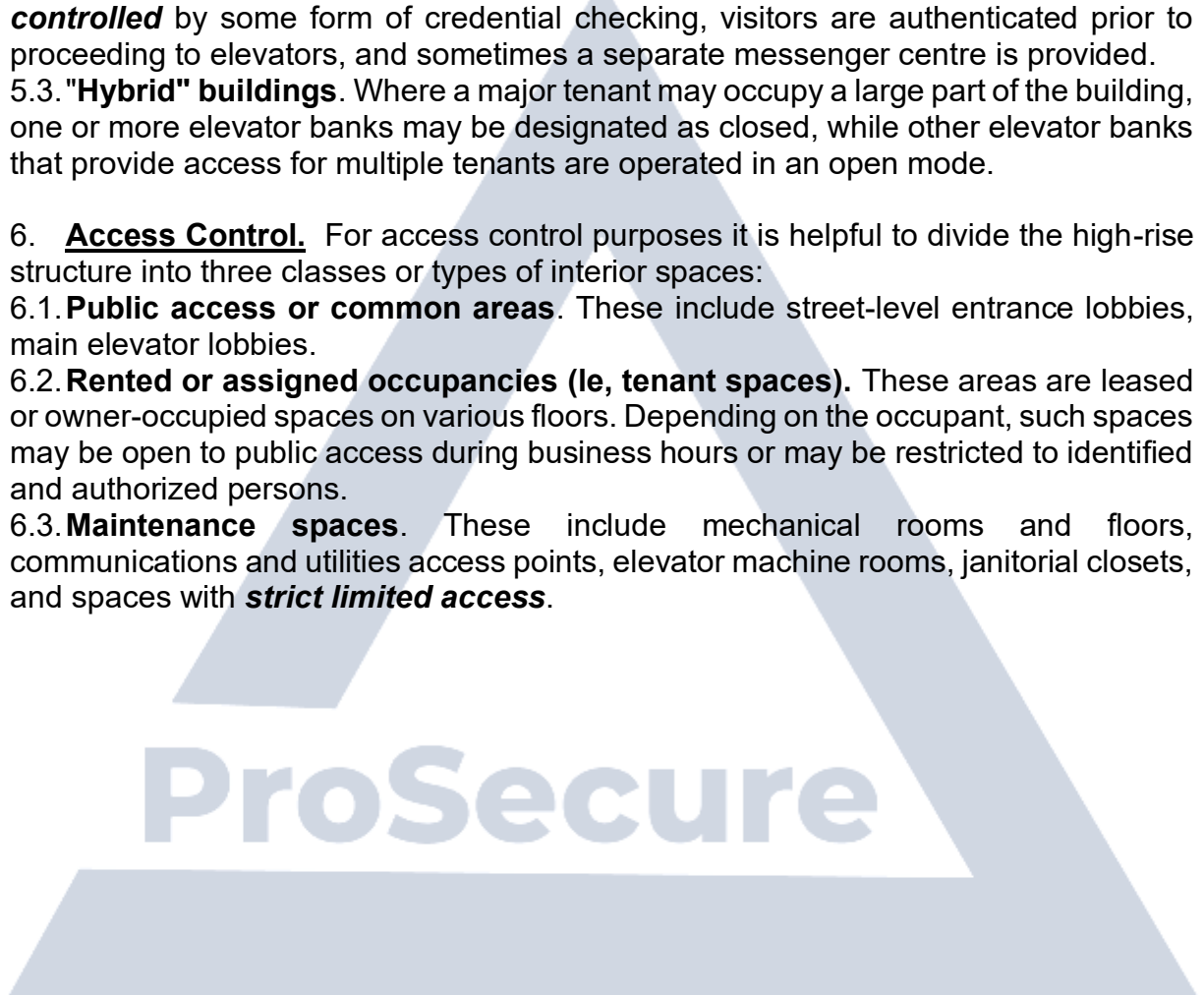
5.3. "**Hybrid**" buildings. Where a major tenant may occupy a large part of the building, one or more elevator banks may be designated as closed, while other elevator banks that provide access for multiple tenants are operated in an open mode.

6. **Access Control.** For access control purposes it is helpful to divide the high-rise structure into three classes or types of interior spaces:

6.1. **Public access or common areas.** These include street-level entrance lobbies, main elevator lobbies.

6.2. **Rented or assigned occupancies (le, tenant spaces).** These areas are leased or owner-occupied spaces on various floors. Depending on the occupant, such spaces may be open to public access during business hours or may be restricted to identified and authorized persons.

6.3. **Maintenance spaces.** These include mechanical rooms and floors, communications and utilities access points, elevator machine rooms, janitorial closets, and spaces with **strict limited access**.

A large, light blue watermark logo is centered on the page. It consists of a large, stylized letter 'A' that serves as a background. Overlaid on the center of the 'A' is the word 'ProSecure' in a bold, sans-serif font. Below the 'A' and the word 'ProSecure' is a horizontal bar, also in the same light blue color, which contains the text 'Process | Performance | Protection' in a smaller, lighter blue font.

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